Probability and Statistics

Probability

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1. Sample space and event
2. Axiom, interpretation, and property of probability
3. Counting technique
4. Conditional probability
5. Independance

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1. Sample space and event
2. Axiom, interpretation, and property of probability
3. Counting technique
4. Conditional probability
5. Independance

Sample space and event

- Sample Space （样本空间）
  - An experiment is any action or process whose outcome is subject to uncertainty
    - Tossing a coin once or several times
  - Sample space, usually denoted by $\mathcal{S}$, is the set of all possible outcomes of that experiment
    - Examining whether a machine is defective or not
      - Possible outcomes: D (defective) and N (not defective)
      - $\mathcal{S} = \{D, N\}$
    - Examining whether two machines are defective or not



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      - Possible outcomes: D (defective) and N (not defective)
      - $\mathcal{S} = \{D, N\}$
    - Examining whether two machines are defective or not
      - $\mathcal{S} = \{DD, DN, ND, NN\}$

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Sample space and event

- Sample Space
  - Tossing one coin: $\mathcal{S} = \{H\ (head), T\ (tail)\}$
  - Tossing two coins: $\mathcal{S} = \{HH, HT, TH, TT\}$
  - Rolling a dice: $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$

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Sample space and event

- Sample Space
  - Two gas stations, each one has 6 gas pumps
  - How many pumps are used in each station?
  - 49 number of possible outcomes

		Second Station						
		0	**1**	**2**	**3**	**4**	**5**	**6**
First Station	**0**	(0, 0)	(0, 1)	(0, 2)	(0, 3)	(0, 4)	(0, 5)	(0, 6)
	1	(1, 0)	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
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Sample space and event

- Sample Space
  - Keep tossing a coin until head side appears
    - $S = \{H, TH, TTH, TTTH, TTTTH, \ldots\}$
    - Infinite number of outcomes

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    - $S = \{H, TH, TTH, TTTH, TTTTH, \ldots\}$
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  - Keep tossing a coin until tail side appears

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Sample space and event

- Sample Space
  - Keep tossing a coin until head side appears
    - $S = \{H, TH, TTH, TTTH, TTTTH, \ldots\}$
    - Infinite number of outcomes
  - Keep tossing a coin until tail side appears
    - $S = \{T, HT, HHT, HHHT, HHHHT, \ldots\}$

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Sample space and event

- Event （事件）
  - Collection of outcomes from a sample space
  - Simple event
    - Event consisting of one outcome
  - Compound event
    - Event consisting of more than one outcomes

Sample space and event

- Event
  - Consider an experiment in which each of three vehicles taking a particular freeway exit turns left(L) or right(R) at the end of the exit ramp
    - Sample space
      - $\{LLL, RLL, LRL, LLR, LRR, RLR, RRL, RRR\}$
    - Simple event
      - All the vehicles turn left
        - $\{LLL\}$
      - All the vehicles turn right
        - $\{RRR\}$
    - Compound event
      - Only one vehicle turns left
        - $\{LRR, RLR, RRL\}$

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Sample space and event

- Simple event
  - There are 49 simple events
  - 2 pumbs are in use in the first station and 4 pumbs in the second station
    - (2, 4)
- Compound event
  - The number of pumps in use are the same
    - {(0, 0) (1, 1) (2, 2) (3, 3) (4, 4) (5, 5) (6, 6)}
  - The total number of pumps in use is four
    - {(0, 4) (1, 3) (2, 2) (3, 1) (4, 0)}
  - At most one pump is in use at each station
    - {(0, 0) (0, 1) (1, 0) (1, 1)}

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Sample space and event

- Keep tossing a coin until head side appears
  - The number of tossing is at most 3



Sample space and event

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    - $\{H, TH, TTH\}$

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Sample space and event

- Keep tossing a coin until head side appears
  - The number of tossing is at most 3
    - $\{H, TH, TTH\}$
  - The number of tossing is even



Sample space and event

- Keep tossing a coin until head side appears
  - The number of tossing is at most 3
    - $\{H, TH, TTH\}$
  - The number of tossing is even
    - $\{TH, TTTH, TTTTTH, TTTTTTTH, \ldots\}$

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Sample space and event

- Event and set theory （事件与集合论）
  - An event is a set, so set theory can be used to manipulate event
  - Union
    - Denoted by A ∪ B, read “A or B”
    - The event consisting of all outcomes that are either in A or in B or in both events
  - Intersection
    - Denoted by A ∩ B, read “A and B”
    - The event consisting of all outcomes that are in both A and B
  - Complement
    - Denoted by $A^{'}$ or $\overline{A}$
    - The event consisting of all outcomes that are not in A

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Sample space and event

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  - An event is a set, so set theory can be used to manipulate event

Sample space and event

- In gas station example, consider the pumps in use in the first gas station
- Let A={0,1,2,3,4}, B={3,4,5,6}, C={1,3,5}
  - A∪B

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  - A ∪ C
    - {0,1,2,3,4,5}
  - A ∩ B

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    - {0,1,2,3,4,5}
  - A ∩ B
    - {3,4}

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Sample space and event

- In gas station example, consider the pumps in use in the first gas station
- Let A={0,1,2,3,4}, B={3,4,5,6}, C={1,3,5}
  - A'
    - {5,6}
  - B'
    - {0,1,2}
  - Let $D = A \cup C$, what is D' ?

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  - A'
    - {5,6}
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  - Let $D = A \cup C$, what is D' ?
    - $D = \{0,1,2,3,4,5\}$, $D' = \{6\}$

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Sample space and event

- Mutually exclusive (disjoint) events （互斥事件）
  - A and B have no outcomes in common
  - $A \cap B = \emptyset$
    - $\emptyset$ denote the null event, which is the event consisting of no outcomes

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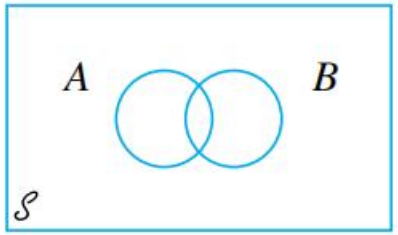
Sample space and event

- Mutually exclusive (disjoint) events
  - Rolling two dices
  - Let event A be 4 appears in the second dice
    - A = {(1, 4), (2, 4), (3, 4), (4, 4), (5, 4), (6, 4)}
  - Let event B be 5 appears in the second dice
    - B = {(1, 5), (2, 5), (3, 5), (4, 5), (5, 5), (6, 5)}
  - $A \cap B = \emptyset$

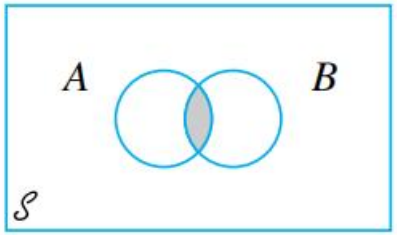
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Sample space and event

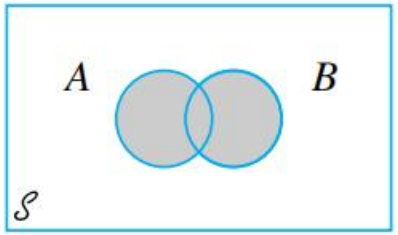
- Venn diagram（维恩图）
  - A pictorial representation of event
  - Draw a rectangle whose interior represent the sample space
  - Any event is represented as the interior of a closed circle



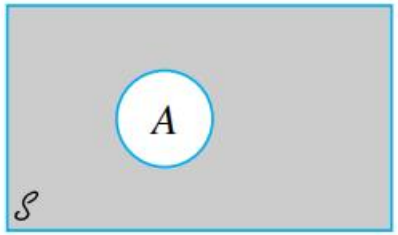
(a) Venn diagram of events A and B



(b) Shaded region is $A \cap B$



(c) Shaded region is $A \cup B$



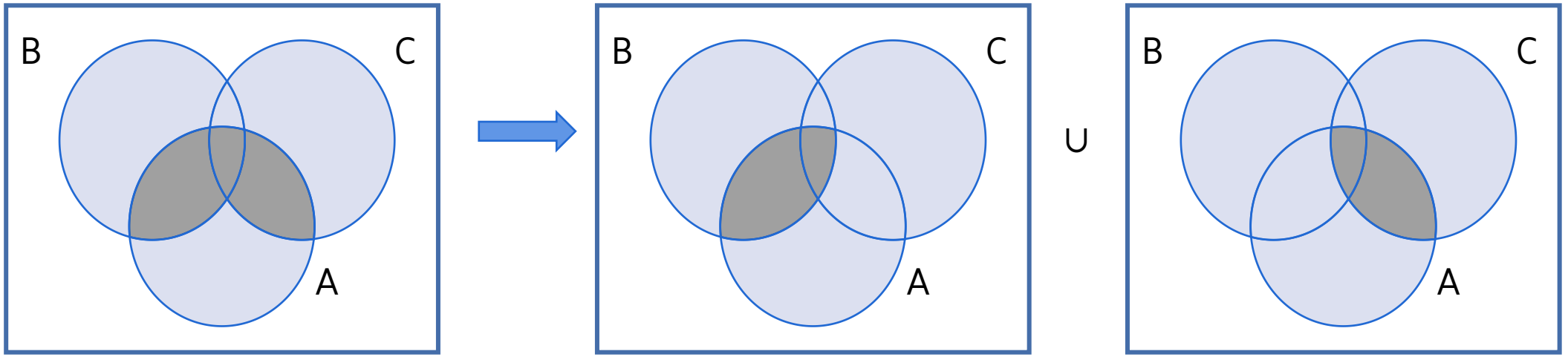
(d) Shaded region is A'



(e) Mutually exclusive events

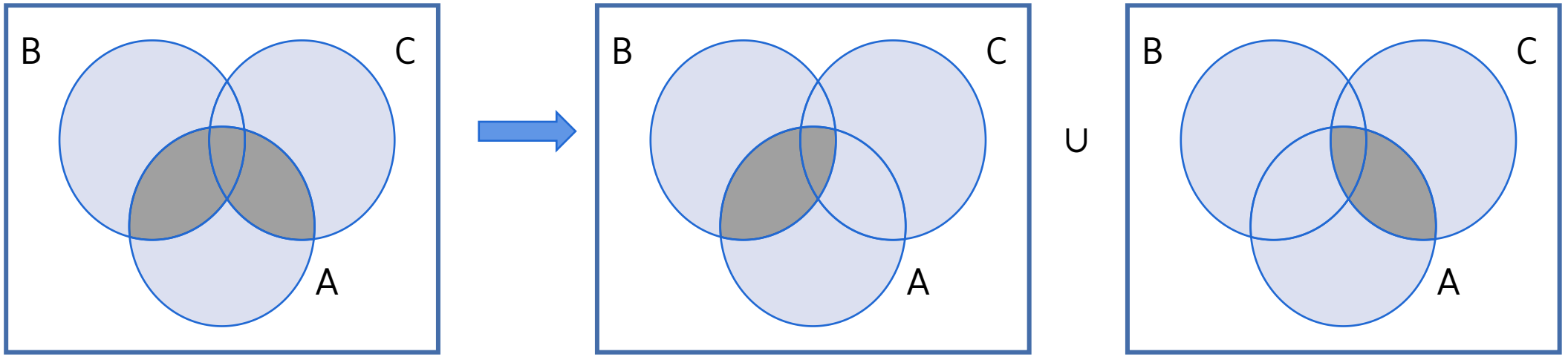
Sample space and event

- Rules of set theory
  - First distributive law
  - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
  - A*(B+C) = (A*B)+(A*C)



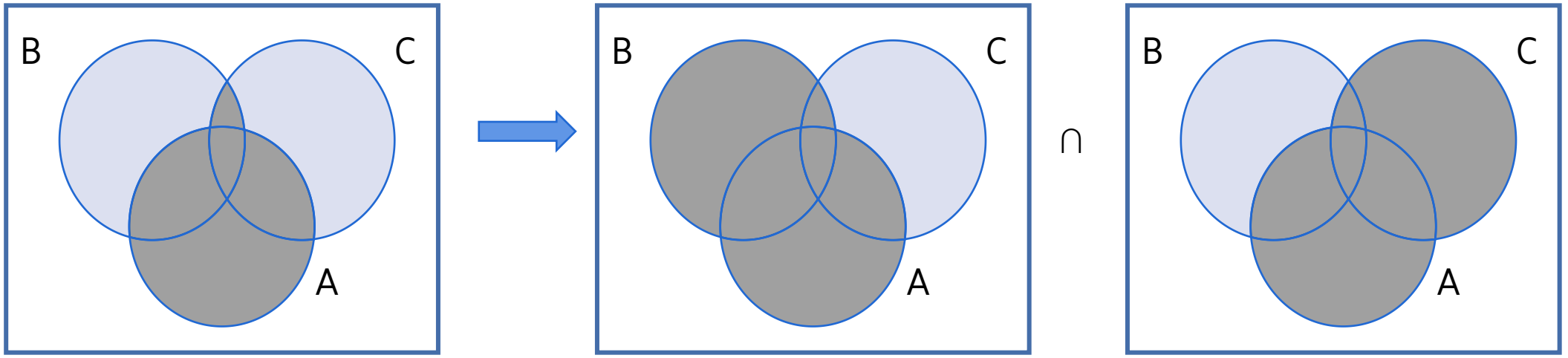
Sample space and event

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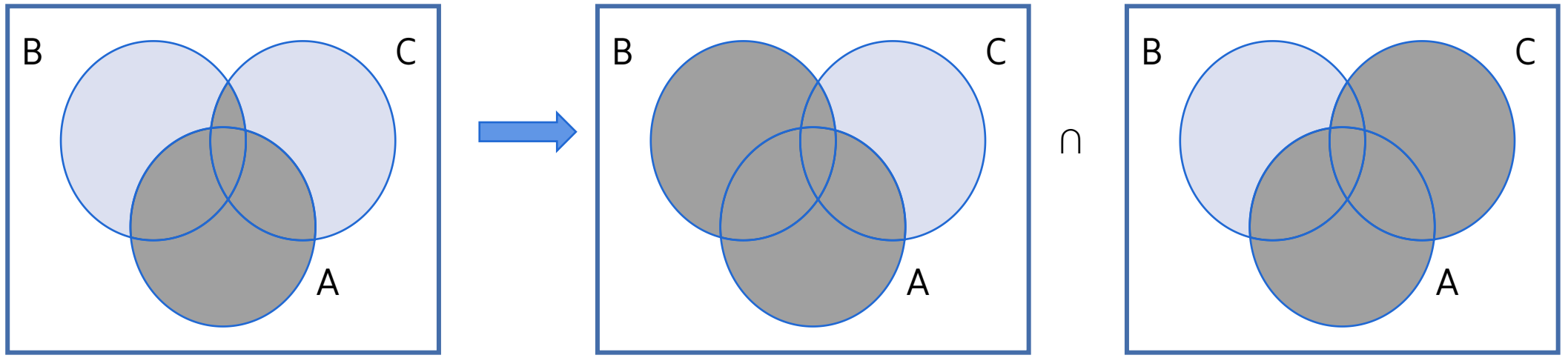
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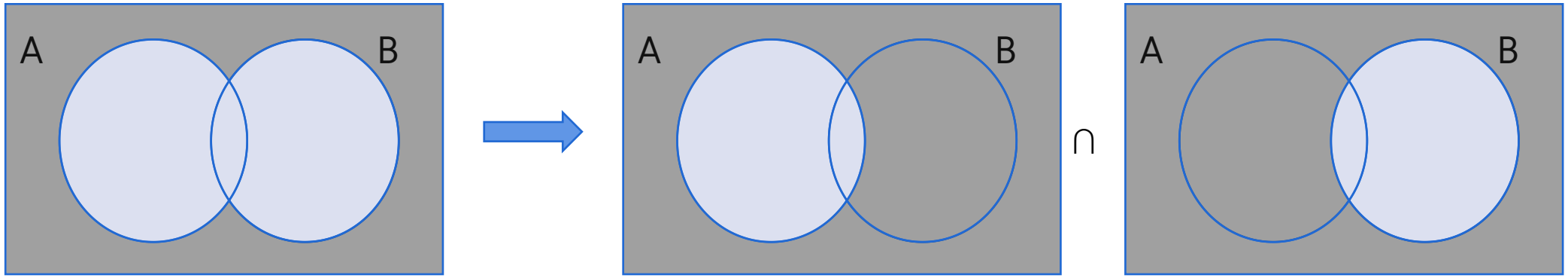
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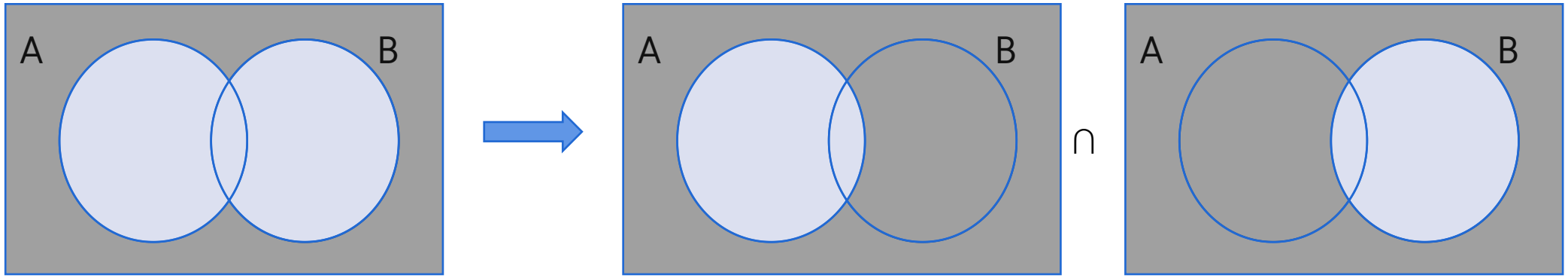
Sample space and event

- Rules of set theory
  - De Morgan`s first law
  - $\overline{A \cup B} = \overline{A} \cap \overline{B}$



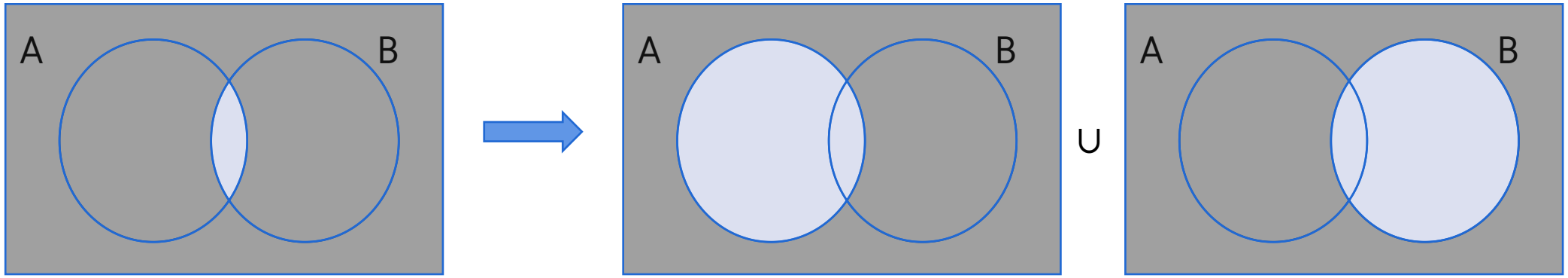
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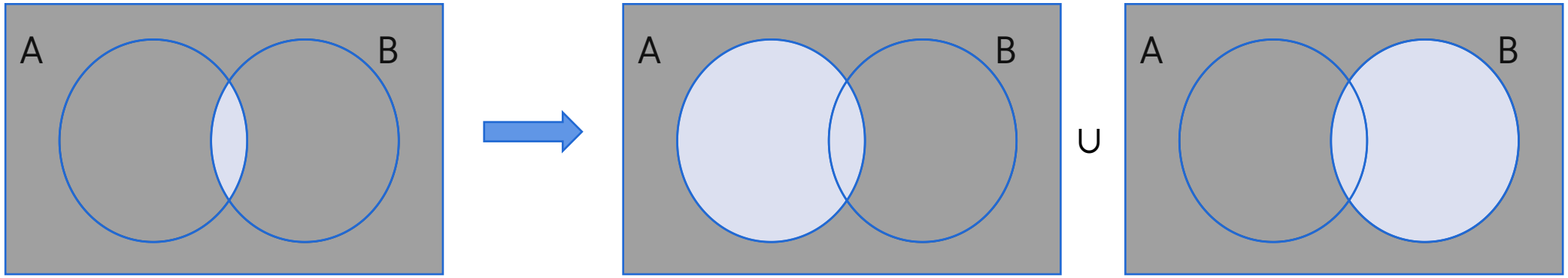
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  - $\overline{A \cap B} = \overline{A} \cup \overline{B}$



1. Sample pace and event
2. Axiom, interpretation, and property of probability
3. Counting technique
4. Conditional probability
5. Independance

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Axiom, interpretation, and property of probability

- Probability
  - The objective of probability is to assign to each event A a value P(A), called the probability of the event A
  - P(A) gives a precise measure of the chance that A will occur
- Axiom of probability（概率公理）
  - The value assignment must satisfy
  - Axiom 1
    - For any event A, $P(A) \geq 0$
  - Axiom 2
    - $P(S) = 1$
  - Axiom 3
    - If A_i, i = 1, 2, 3, ..., is a finite or infinite collection of disjoint events
    - $P(\cup A_i) = \sum P(A_i)$

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Axiom, interpretation, and property of probability

- Examining whether a machine is defective or not
  - Possible outcomes: D (defective) and N (not defective)
  - $\mathcal{S} = \{\mathrm{D}, \mathrm{N}\}$
  - Suppose $\mathrm{P}(\mathrm{D}) = 0.2$ (Axiom 1)
  - Because $\mathrm{D} \cap \mathrm{N} = \emptyset$, $\mathrm{P}(\mathrm{D} \cup \mathrm{N}) = \mathrm{P}(\mathrm{D}) + \mathrm{P}(\mathrm{N})$ (Axiom 3)
  - Because $\mathrm{P}(\mathcal{S}) = \mathrm{P}(\mathrm{D} \cup \mathrm{N}) = \mathrm{P}(\mathrm{D}) + \mathrm{P}(\mathrm{N}) = 1$ (Axiom 2)
  - $\mathrm{P}(\mathrm{N}) = 1 - \mathrm{P}(\mathrm{D}) = 0.8$

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Axiom, interpretation, and property of probability

- A worker continuously checks products coming from a product line until a defective one is found
  - Let E_i be the event that *i* number of products are checked
  - $E_1 = \{D\}, E_2 = \{N, D\}, E_3 = \{N, N, D\}$ ...
  - $E_i \cap E_j = \emptyset,\ i \neq j$
  - $\mathcal{S} = \{ \bigcup_i^{\infty} E_i \}$
  - $P(\mathcal{S}) = 1$

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Axiom, interpretation, and property of probability

- A worker continuously checks products coming from a product line until a defective one is found
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  - $E_i \cap E_j = \emptyset,\ i \neq j$
  - $\mathcal{S} = \{ \bigcup_i^\infty E_i \}$
  - $P(\mathcal{S}) = 1$
    - $P(D) = 0.2, P(N) = 0.8$
    - $P(E_i) = 0.8^{i-1} * 0.2$ (independance)
    - $P(E_3) = P(N, N, D) = 0.8^2 * 0.2$
    - $P(\mathcal{S}) = P(\bigcup_i^\infty E_i) = \sum_{i=1}^\infty P(E_i) = \sum_{i=1}^\infty 0.8^{i-1} * 0.2 = 0.2 * \sum_{i=0}^\infty 0.8^i$
    - $\sum_{i=0}^\infty 0.8^i = 1 / 0.2$
    - $P(\mathcal{S}) = 0.2 * 1 / 0.2 = 1$

Axiom, interpretation, and property of probability

- Probability
  - Relative frequency in sample data set



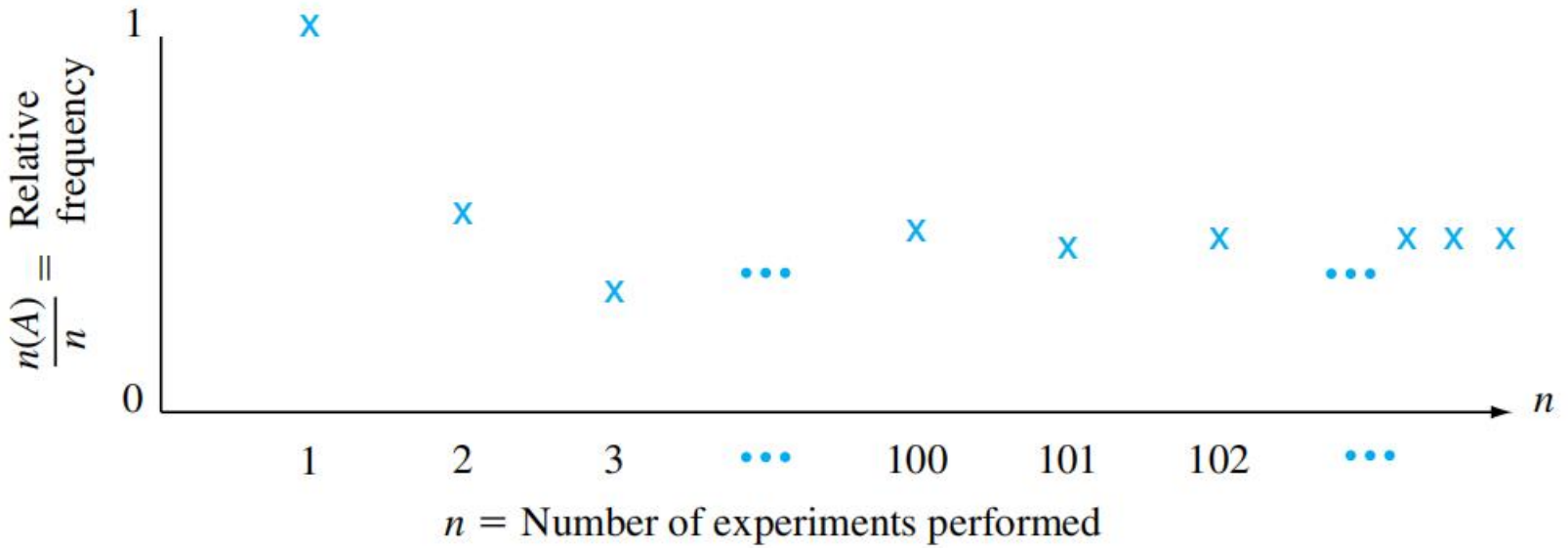
Axiom, interpretation, and property of probability

- Probability
  - Relative frequency in sample data set
    - $\frac{Freqency}{Number\ of\ sample\ data\ set} = \frac{Number\ of\ times\ a\ value\ appears}{Number\ of\ sample\ data\ set}$
  - Consider an experiment that can be repeatedly performed, and let A be an event consisting of a fixed set of outcomes
    - Repeatedly toss a coin, event A can be the head side appears
  - Let n be the number of repeated experiments, and $n(A)$ the number of times A occurs
  - Relative frequency of A $= \frac{Number\ of\ times\ A\ occurs\ in\ experiments}{Number\ of\ experiments} = \frac{n(A)}{n}$

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Axiom, interpretation, and property of probability

- Limiting relative frequency
  - As the number of experiments grows large, relative frequency approaches a limiting value, which is called limiting relative frequency



Axiom, interpretation, and property of probability

- Limiting relative frequency
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	$n = 5$		$n = 50$		$n = 500$	
	n_H	f	n_H	f	n_H	f
A	2	0.4	22	0.44	251	0.502
B	3	0.6	25	0.50	249	0.498
C	1	0.2	21	0.42	256	0.512
D	5	1.0	25	0.50	247	0.494

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Axiom, interpretation, and property of probability

- Limiting relative frequency
  - As the number of experiments grows large, relative frequency approaches a limiting value, which is called limiting relative frequency

	n	n_H	f
A	2048	1061	0.5181
B	4040	2048	0.5069
C	12000	6019	0.5016
D	24000	12012	0.5005

Axiom, interpretation, and property of probability

- Probability
  - The probability of an event A is its limiting relative frequency
  - $P(A) = \lim_{n\to\infty} \frac{n(A)}{n}$
  - This relative frequency interpretation of probability is said to be objective because it is based on property of experiment rather than on any particular individual
  - Two different observers of a sequence of coin tosses should both use the same probability assignments since the observers have nothing to do with limiting relative frequency

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Axiom, interpretation, and property of probability

- Probability
  - Because the objective interpretation of probability is based on the notion of limiting frequency, its applicability is limited to experimental situations that are repeatable
  - But the language of probability is often used in connection with situations that
    - It is likely that our company will be awarded the contract
    - The chances are good for a peace agreement
    - The probability is 0.9 that we will get the contract
  - Different observers may have different prior information and opinions, so the assignment of probability is subjective
  - In this course, we will focus on objective probability

Axiom, interpretation, and property of probability

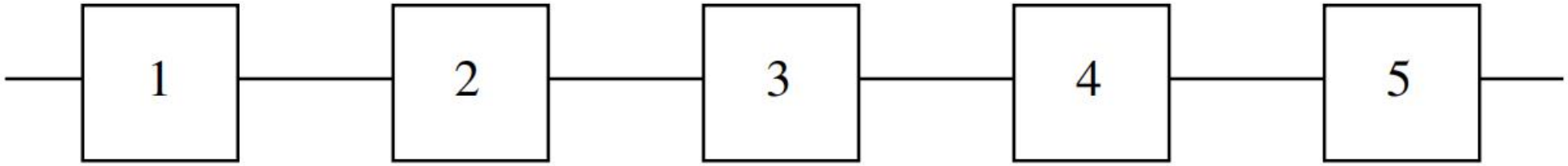
- Property of probability

For any event A, $P(A) + P(A') = 1$, from which $P(A) = 1 - P(A')$.

Proof In Axiom 3, let $k = 2$, $A_1 = A$, and $A_2 = A'$. Since by definition of A', $A \cup A' = \mathcal{S}$ while A and A' are disjoint, $1 = P(\mathcal{S}) = P(A \cup A') = P(A) + P(A')$.

Axiom, interpretation, and property of probability

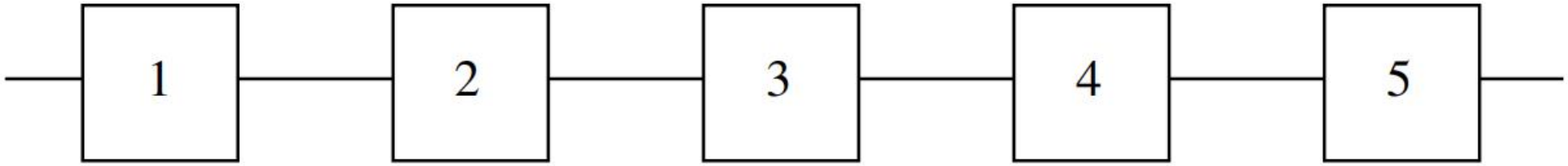
- Property of probability
  - Consider five identical components connected in series
  - The system fails if at least one component fails
  - Let event A be the system fails



- If the probability that a component fails and does not fail is 0.1 and 0.9, what is P(A)?

Axiom, interpretation, and property of probability

- Property of probability
  - Consider five identical components connected in series
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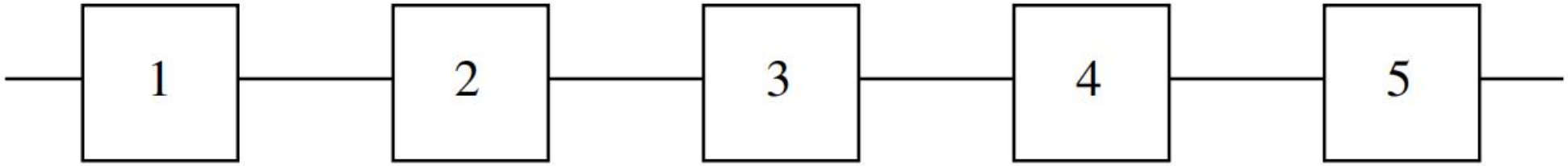


- If the probability that a component fails and not is 0.1 and 0.9, what is P(A)?
  - A = {SSSSF, SSFFS, FSSSS ...}, totally 31 outcomes in A!



Axiom, interpretation, and property of probability

- Property of probability
  - Consider five identical components connected in series
  - The system fails if at least one component fails
  - Let event A be the system fails



- If the probability that a component fails and not is 0.1 and 0.9, what is P(A)?
  - A = {SSSSF, SSFFS, FSSSS ...}, totally 31 outcomes in A!
  - $P(A') = P(SSSSS) = 0.9^5 = 0.59 \Rightarrow P(A) = 1 - P(A') = 0.41$

Axiom, interpretation, and property of probability

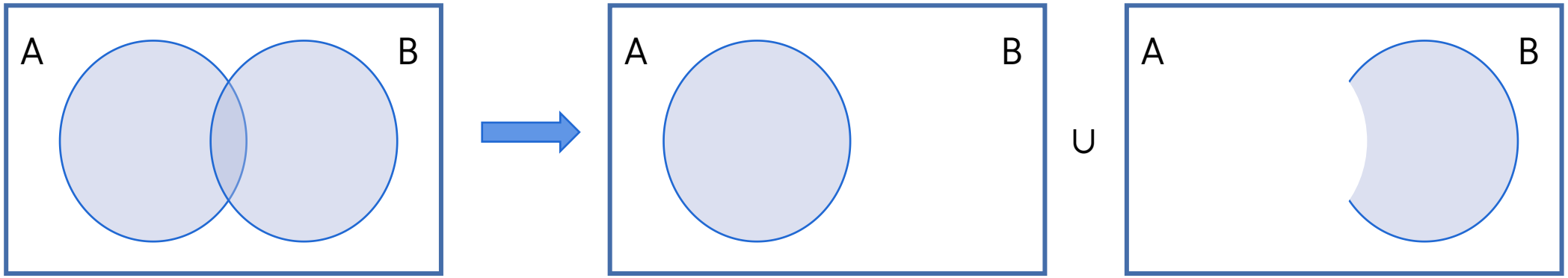
- Property of probability

For any event A, $P(A) \leq 1$.

This is because $1 = P(A) + P(A') \geq P(A)$ since $P(A') \geq 0$.

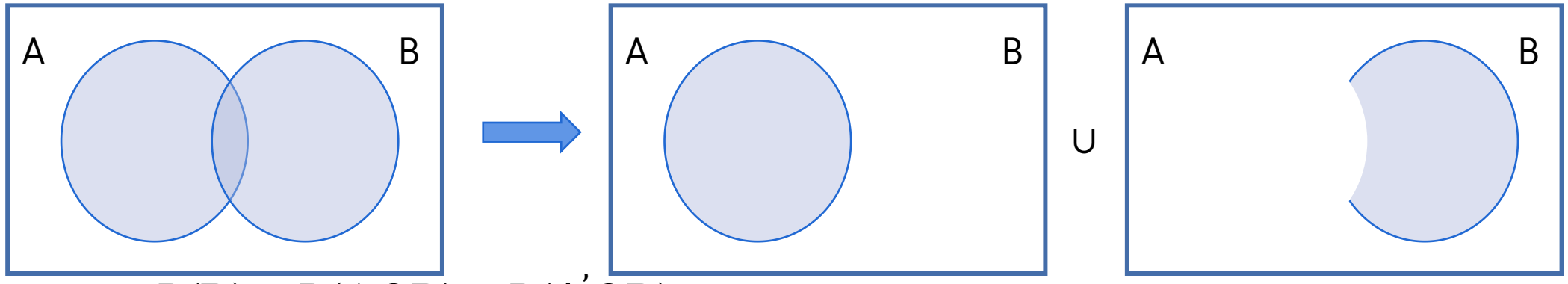
Axiom, interpretation, and property of probability

- Property of probability
  - For any two events A and B, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
  - Proof:
    - $P(A \cup B) = P(A) + P(A' \cap B)$

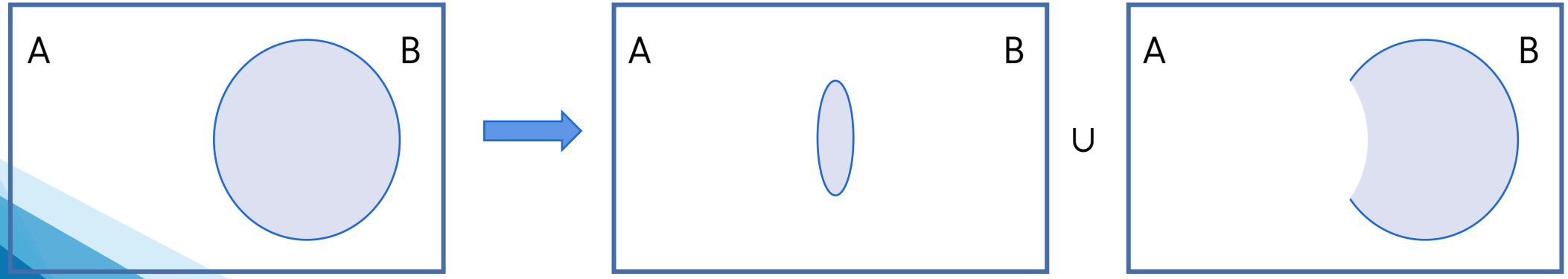


Axiom, interpretation, and property of probability

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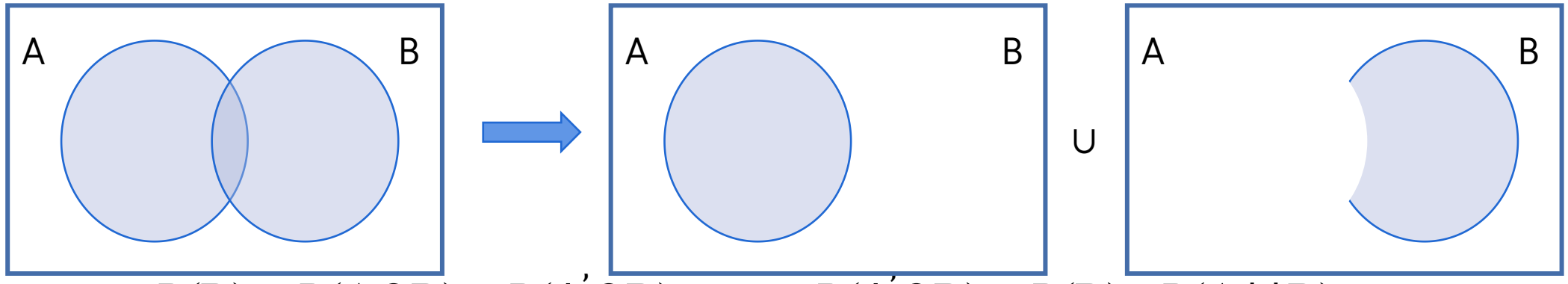


    - $P(B) = P(A \cap B) + P(A' \cap B)$

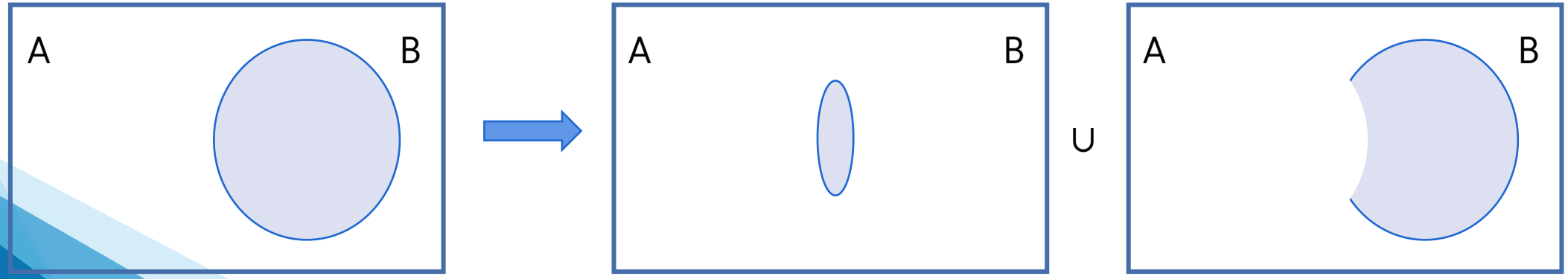


Axiom, interpretation, and property of probability

- Property of probability
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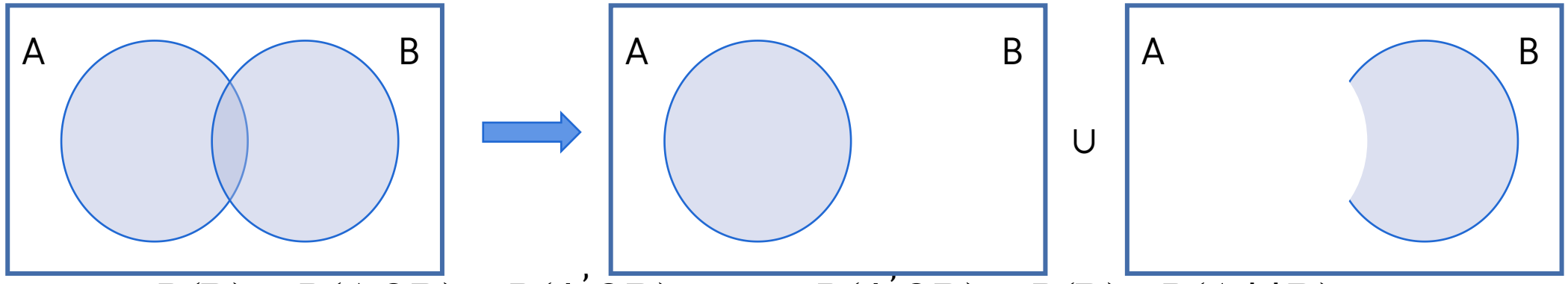


- $P(B) = P(A \cap B) + P(A' \cap B) \Rightarrow P(A' \cap B) = P(B) - P(A \cup B)$

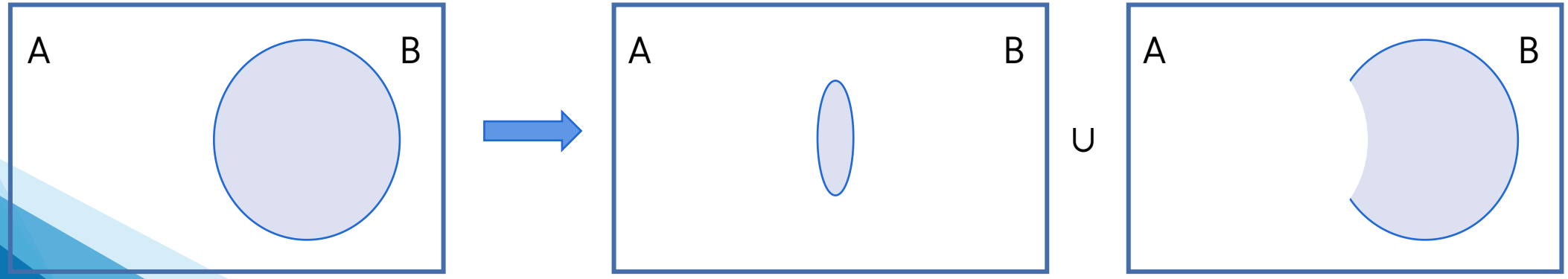


Axiom, interpretation, and property of probability

- Property of probability
  - For any two events A and B, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
  - Proof:
    - $P(A \cup B) = P(A) + P(A' \cap B)$ ⟶ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



    - $P(B) = P(A \cap B) + P(A' \cap B)$ ⟶ $P(A' \cap B) = P(B) - P(A \cup B)$



Axiom, interpretation, and property of probability

- 60% of the students take C++ course, and 80% of the students take Java course, 50% of the students take both courses
- What is the probability for a randomly selected student to take at least one of the courses?

Axiom, interpretation, and property of probability

- 60% of the students take C++ course, and 80% of the students take Java course, 50% of the students take both courses
- What is the probability for a randomly selected student to take at least one of the courses?
  - Let A = {the student takes C++ course}, B = {the student takes Java course}, C = {the student takes both of C++ and Java courses}
  - $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 60\% + 80\% - 50\% = 0.9$

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Axiom, interpretation, and property of probability

- 60% of the students take C++ course, and 80% of the students take Java course, 50% of the students take both courses
- What is the probability for a randomly selected student to take at least one of the courses?
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- What is the probability for the student to take only Java course?

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Axiom, interpretation, and property of probability

- 60% of the students take C++ course, and 80% of the students take Java course, 50% of the students take both courses
- What is the probability for a randomly selected student to take at least one of the courses?
  - Let A = {the student takes C++ course}, B = {the student takes Java course}, C = {the student takes both of C++ and Java courses}
  - $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 60\% + 80\% - 50\% = 0.9$
- What is the probability for the student to take only Java course?
  - $P(A' \cap B) = P(A \cup B) - P(A) = 0.9 - 0.6 = 0.3$

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Axiom, interpretation, and property of probability

- 60% of the students take C++ course, and 80% of the students take Java course, 50% of the students take both courses
- What is the probability for a randomly selected student to take at least one of the courses?
  - Let A = {the student takes C++ course}, B = {the student takes Java course}, C = {the student takes both of C++ and Java courses}
  - $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 60\% + 80\% - 50\% = 0.9$
- What is the probability for the student to take only Java course?
  - $P(A' \cap B) = P(A \cup B) - P(A) = 0.9 - 0.6 = 0.3$
- What is the probability to take only C++ course?

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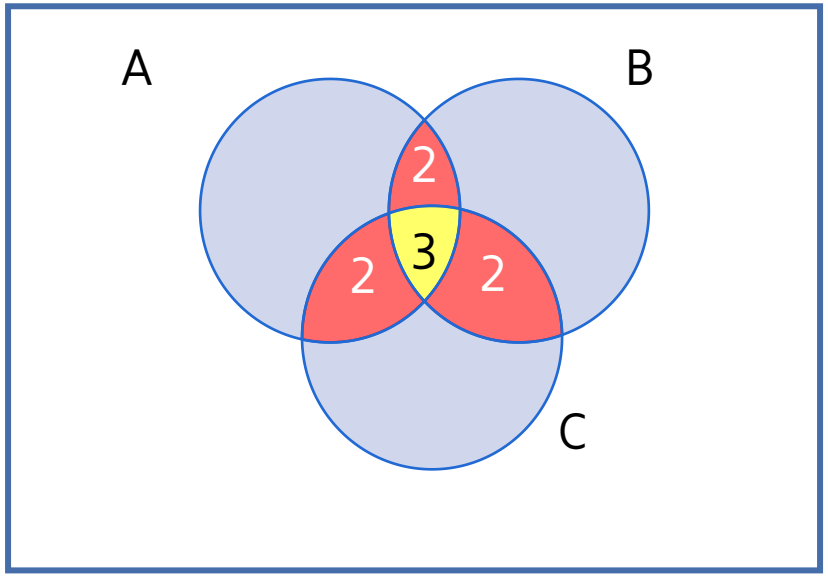
Axiom, interpretation, and property of probability

- 60% of the students take C++ course, and 80% of the students take Java course, 50% of the students take both courses
- What is the probability for a randomly selected student to take at least one of the courses?
  - Let A = {the student takes C++ course}, B = {the student takes Java course}, C = {the student takes both of C++ and Java courses}
  - $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 60\% + 80\% - 50\% = 0.9$
- What is the probability for the student to take only Java course?
  - $P(A' \cap B) = P(A \cup B) - P(A) = 0.9 - 0.6 = 0.3$
- What is the probability to take only C++ course?
  - $P(A \cap B') = P(A \cup B) - P(B) = 0.9 - 0.8 = 0.1$

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Axiom, interpretation, and property of probability

- Union property for more than two events
  - Given three events A, B, and C
  - $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$



Axiom, interpretation, and property of probability

- Exercises
  - A small country has a population of 17 million people of whom 8.4 million are male and 8.6 million are female. If 75% of the male population and 63% of the female population are literate, what percentage of the total population is literate?

Axiom, interpretation, and property of probability

- Exercises
  - A small country has a population of 17 million people of whom 8.4 million are male and 8.6 million are female. If 75% of the male population and 63% of the female population are literate, what percentage of the total population is literate?
  - Total number of literate population = 8.4 * 75% + 8.6 * 63% = 11.7
  - The percentage of the total population is = $\frac{11.7}{17}$ * 100 = 68.93%

Axiom, interpretation, and property of probability

- Exercises
  - Two events A and B have the following probabilities: P(A) = 0.6, P(B) = 0.7, and P(A ∩ B) = p. Find the range of values that p can take

Axiom, interpretation, and property of probability

- Exercises
  - Two events A and B have the following probabilities: $P(A) = 0.6$, $P(B) = 0.7$, and $P(A \cap B) = p$. Find the range of values that p can take
  - $P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq 1$
  - $P(A) + P(B) - 1 \leq P(A \cap B)$ ⟶ $P(A \cap B) \geq 0.3$

Axiom, interpretation, and property of probability

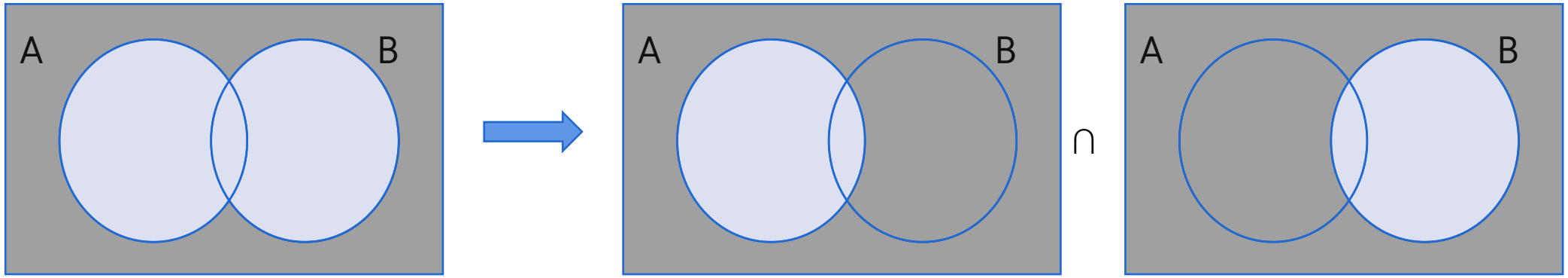
- Exercises
  - Two events A and B have the following probabilities: P(A) = 0.5, P(B) = 0.6, and $P(\overline{A} \cap \overline{B}) = 0.25$. Find the value of $P(A \cap B)$

Axiom, interpretation, and property of probability

- Exercises
  - Two events A and B have the following probabilities: P(A) = 0.5, P(B) = 0.6, and $P(\overline{A} \cap \overline{B}) = 0.25$. Find the value of $P(A \cap B)$
  - De Morgan's first law

Sample space and event

- Rules of set theory
  - De Morgan`s first law
  - $\overline{A \cup B} = \overline{A} \cap \overline{B}$



Axiom, interpretation, and property of probability

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  - De Morgan's first law
  - $P(\overline{A} \cap \overline{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 0.25 \Rightarrow P(A \cup B) = 0.75$
  - $P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow P(A \cap B) = 0.35$

Axiom, interpretation, and property of probability

- Exercises
  - Let A be the set of positive even integers, let B be the set of positive integers that are divisible by 3, and let C be the set of positive odd integers, describe the following sets:
    - $A \cup B$

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    - $A \cup (B \cap C)$: even positive integers or odd positive integers that are divisible by 3

1. Sample pace and event
2. Axiom, interpretation, and property of probability
3. Counting technique
4. Conditional probability
5. Independance

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Counting technique

- Equally likely event
  - When the outcomes of an experiment are equally likely, the task of computing probabilities reduces to counting
  - P(A) = $\frac{n(A)}{n}$, n is the number of outcomes in the sample space, n(A) is the number of outcomes in event A
  - What if n is very large? It is difficult to list all the outcomes

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- Counting technique
  - Product rule （乘法原理）
  - Permutation （排列）
  - Combination （组合）

Counting technique

- Product rule
  - Ordered pair （有序对）
    - If A and B are two objects, then (A, B) ≠ (B, A)
  - Couting the number of ordered pair
    - If the first element of object of an ordered pair can be selected in n_1 ways, and for each of these n_1 ways the second element of the pair can be selected in n_2 ways, then the number of ordered pairs is $n_1 * n_2$

Counting technique

- Product rule
  - If the number of male students is 15, and the number of female students is 23, how many pair of male and female students can be made?

Counting technique

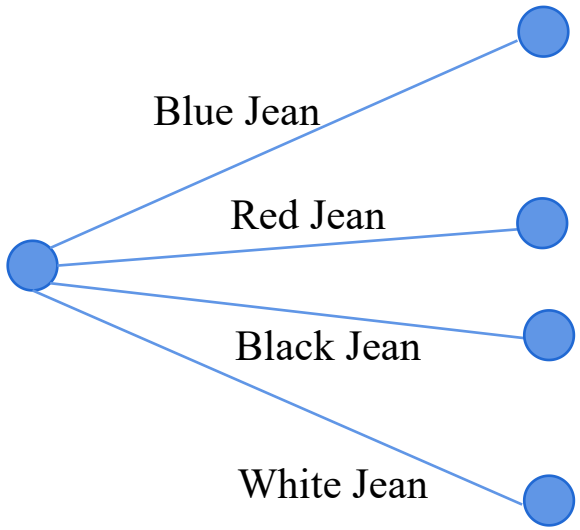
- Product rule
  - If the number of male students is 15, and the number of female students is 23, how many pair of male and female students can be made?
  - 15 * 23 = 345
  - If four students wear blue, red, black, white jeans, and two students wear white and blue T shirts, how many different pairs of students with jean and T shirt can be made?

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  - If four students wear blue, red, black, white jeans, and two students wear white and blue T shirts, how many different pairs of students with jean and T shirt can be made?
  - 4 * 2 = 8

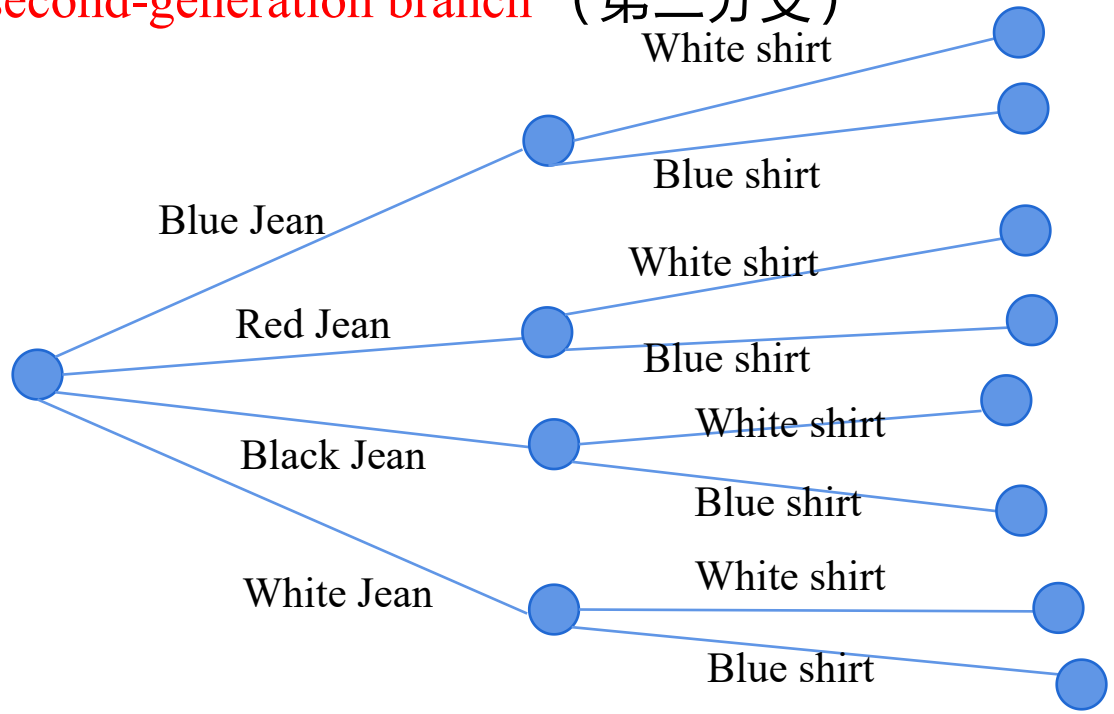
Counting technique

- Tree diagram （树形图）
  - Represent pictorially all the possibilities of ordered pairs
  - Starting from a point on the left side, for each possible first element of a pair a straight-line segment emanates rightward, each of these lines is referred to as first-generation branch



Counting technique

- Tree diagram
  - Represent pictorially all the possibilities of ordered paris
  - Starting from a point on the left side, for each possible first element of a pair a straight-line segment emanates rightward, each of these lines is referred to as first-generation branch （第一分支）
  - Draw another line segment for each possible choice of a second element of the pair, each line is referred to as second-generation branch （第二分支）



Counting technique

- K-tuple （K元组）
  - An ordered collection of k objects
  - Product rule for K-tuple
    - There are n_1 possible choices for the first element, for each choice of the first element, n_2 posssible choices can be made for the second element, for each choice of the second element, n_3 possible choices can be made for the third element, and so on so forth
    - The number of possible K-tuples is $\prod_{i=1}^{K} n_i$

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Counting technique

- Permutation （排列）
  - An ordered subset
  - The number of permutations of size k that can be formed from n individuals is denoted by $P_{k,n}$
  - $P_{k,n} = n * (n-1) * (n-2) \ldots (n - (k-2)) * (n-(k-1)) = \frac{n!}{(n-k)!}$
  - n! is called n factorial (阶乘), $n! = n * (n-1) * (n-2) \ldots * 1$
  - $P_{2,3} = 3 * 2 = \frac{3!}{(3-2)!}$
  - Note the relationship with product rule

Counting technique

- Permutation
  - There are 10 teaching assistants available for grading exam in a particular course The exam consists of 4 questions
  - In how many ways can assistants be chosen to grade the exam?

Counting technique

- Permutation
  - There are 10 teaching assistants available for grading exam in a particular course. The exam consists of 4 questions
  - In how many ways can assistants be chosen to grade the exam?
  - n = the number of assistants = 10
  - k = the number of questions = 4
  - The number of grading assignments is $P_{k,n}$=10 * 9 * 8 * 7 = 5040

Counting technique

- Permutation
  - Birthday paradox (生日悖论)
  - What is the probability that at least two students in the class of k students have the same birthday?
  - Let A = {all of the k students have different birthdays}
  - $P_{k,365} = \frac{365!}{(365-k)!}$
  - $P(A) = \frac{P_{k,365}}{365^k}$
  - The probability that at least two students have the same birthday = 1 - P(A)

1	2	3	4	5	10
0.00	0.27	0.82	1.64	2.71	11.69
15	20	30	40	50	100
25.29	41.14	70.63	89.12	97.04	99.99

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Counting technique

- Permutation
  - Birthday paradox
  - For 50 students, the probability of at least two students sharing the same birthday is around 97%, it is counter-intuitive

Counting technique

- Permutation
  - Birthday paradox
  - For 50 students, the probability of at least two students sharing the same birthday is around 97%, it is counter-intuitive
  - There are many pairs in 50 students
    - Even if the probability for two students to have identical birthday is small
    - There are 1225 possible pairs to compare
  - Personal experience
    - In everyday life, you might not often come across someone with the same birthday as you, reinforcing the belief that matching birthdays are rare

Counting technique

- Combination （组合）
  - An unordered subset
  - The number of combinations of size k that can be formed from n individuals is denoted by $C_{k,n}$
  - $C_{k,n} = \frac{P_{k,n}}{k!} = \frac{n!}{k! * (n-k)!}$



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  - $C_{k,n} = \frac{P_{k,n}}{k!} = \frac{n!}{k! * (n-k)!}$
  - $P_{k,n} = k! * C_{k,n}$, because the number of ways to select k individuals from n individuals is $C_{k,n}$, and for each of these combinations the number of possible ordering is $k!$
  - $P_{k,n} = k! * C_{k,n} \longrightarrow C_{k,n} = \frac{P_{k,n}}{k!} = \frac{n!}{k! * (n-k)!}$

Counting technique

- Couting technique application
  - Four married couples bought tickets for eight seats in a row for a football game

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    - Since there are 4 couples and each couple can be arranged in only one way (husband to the left of his wife)
    - $P_{4,4} = 4!$

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    - $P_{4,4} * {P_{2,2}}^4 = P_{4,4} * (2)^4$

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Counting technique

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  - There are N products, D of them are defective
  - What is the probability that k defective products are selected if m products are randomly selected (so the number of defective ones is k, and the number of intact ones is m - k)?



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  - $$\frac{Number\ of\ ways\ that\ k\ defective\ ones\ selected * Number\ of\ ways\ m-k\ intact\ ones\ selected}{Number\ of\ ways\ m\ products\ are\ selected}$$
    - The number of combinations of picking m products is $C_{m,N}$
    - The number of combinations of picking k defective products is $C_{k,D}$
    - The number intact products in m selected products is m - k, the number of intact products in the total products is N - D
    - Hence, the number of ways to select m - k intact products from the total N - D intact products is $C_{m-k,N-D}$
    - Hence, $\frac{C_{k,D} * C_{m-k,N-D}}{C_{m,N}}$

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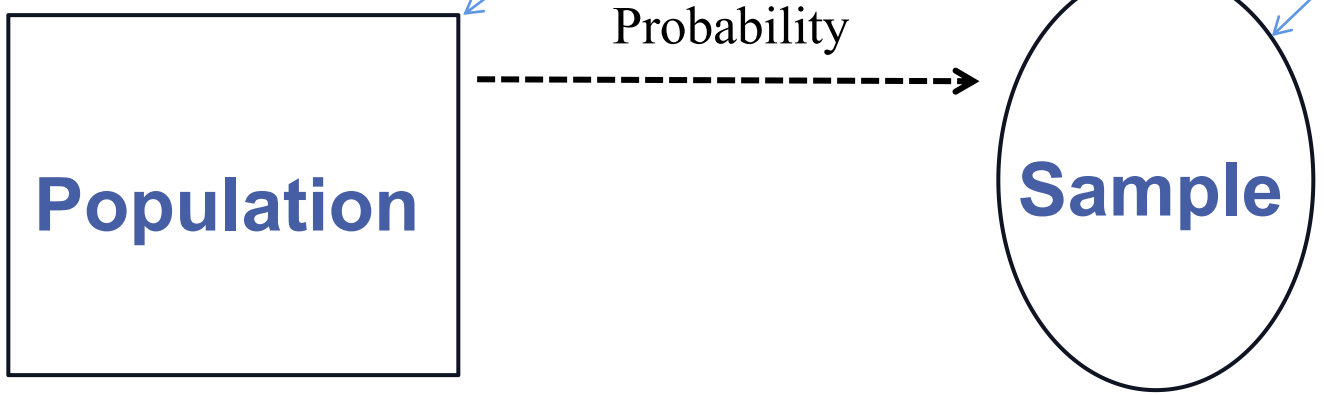
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Counting technique

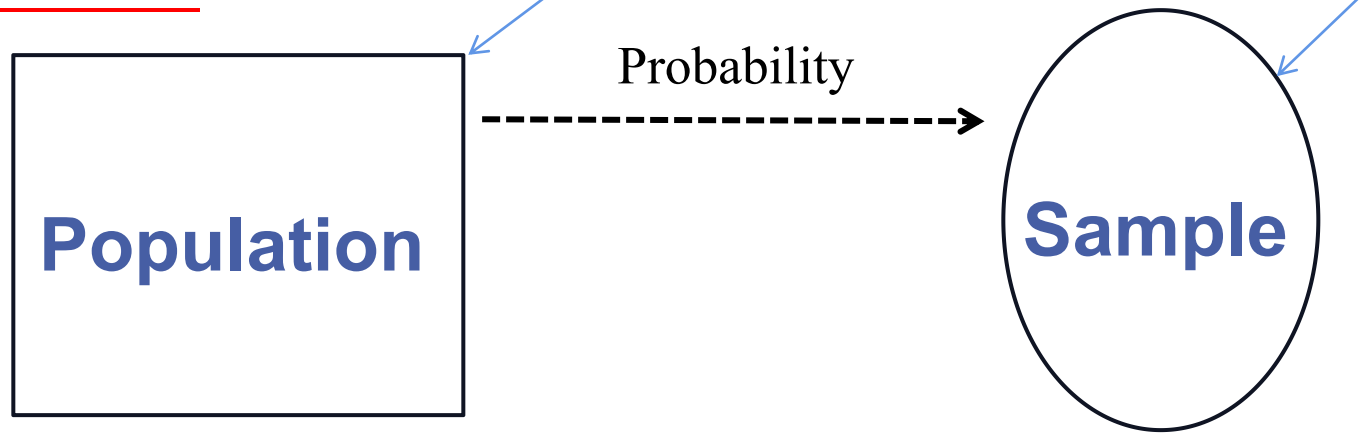
- The relationship between statistics and probability
  - The number of male students in a class of 50 students is 10
  - What is the probability for 5 male students to be selected if 30 students are selected randomly?





Counting technique

- The relationship between statistics and probability
  - The number of male students in a class of 50 students is 10
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  - The number of combinations of choosing 30 students is $C_{30,50}$
  - The number of combinations of choosing 5 male students in 30 students is $C_{5,10} * C_{25,40}$
  - Hence, $\frac{C_{5,10} * C_{25,40}}{C_{30,50}} = 0.21$

1. Sample pace and event
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5. Independance

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Conditional probability

- Conditional probability (条件概率)
  - The probability for an event to occur given that another event has already occurred
  - It helps in updating the probability of an event based on additional information or evidence
  - The conditional probability means the probability that event A will occur if we know that event B has occured already, denoted by P(A|B)

Conditional probability

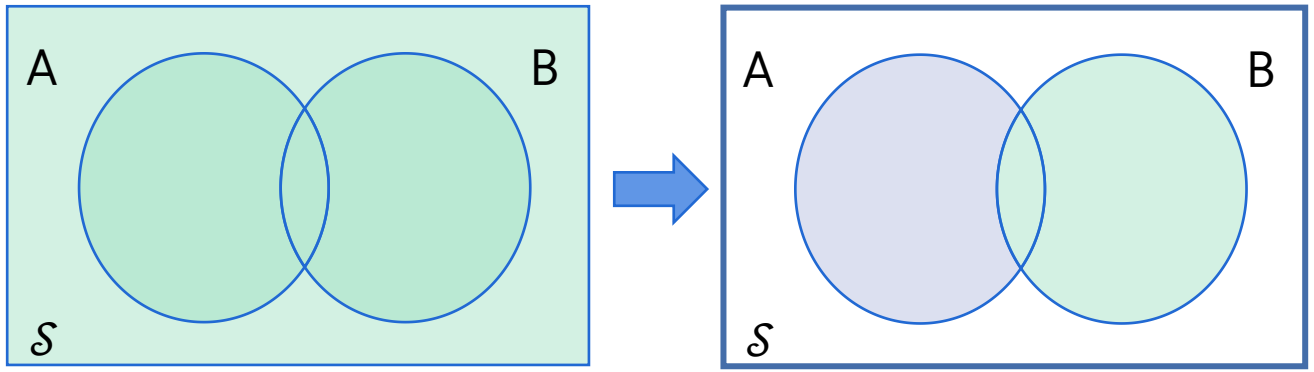
- Conditional probability (条件概率)
  - A is the event that an individual having a particular disease
  - B is the event of negative blood test
  - The probability that the individual has the disease if his blood test is negative is P(A|B)

Conditional probability

- Conditional probability
  - For any two events A and B, the conditional probability of A given that B has occurred can be obtained by
  - $P(A|B) = \frac{P(A \cap B)}{P(B)}$

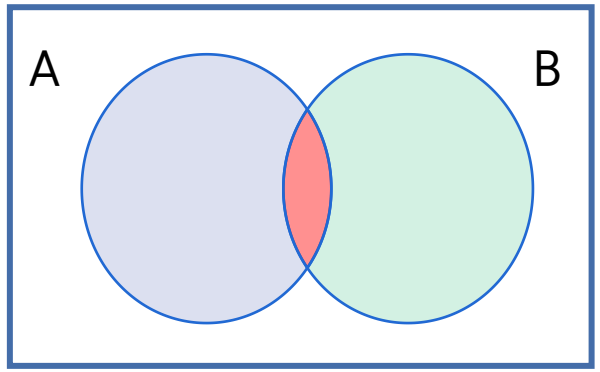
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  - A can occur if and only if one of the outcomes in the intersection occurs



- Conditional probability
  - Suppose that 50% of all the individuals bought football, 70% bought basketball, and 30% bought both football and basketball
  - What is the probability for an individual to buy a football if it is found that he already bought a basketball?

- Conditional probability
  - Suppose that 50% of all the individuals bought football, 70% bought basketball, and 30% bought both football and basketball
  - What is the probability for an individual to buy a football if it is found that he already bought a basketball?
  - A = {he bought a football}, B = {he bought a basketball}, C = {he bought both}
  - $P(A|B) = P(A \cap B) / P(B) = 30\% / 70\% = 42\%$

Conditional probability

- Conditional probability
  - The probability that a married man votes is 0.45, the probability that a married woman votes is 0.40, and the probability that a married woman votes given that her husband does is 0.60
  - What is the probability that a man votes given that his wife does?

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Conditional probability

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  - The probability that a married man votes is 0.45, the probability that a married woman votes is 0.40, and the probability that a married woman votes given that her husband does is 0.60
  - What is the probability that a man votes given that his wife does?
  - A = {husband votes}, B = {wife votes}
  - $P(A) = 0.45$, $P(B) = 0.4$, $P(B|A) = 0.6$
  - $P(B|A) = P(A \cap B) / P(A) \rightarrow P(A \cap B) = P(B|A) * P(A)$ (multipication rule of conditional probability)
  - $P(A \cap B) = P(B|A) * P(A) = 0.6 * 0.45 = 0.27$
  - $P(A|B) = P(A \cap B) / P(B) = 0.27 / 0.40 = 0.67$

- Conditional probability
  - If one rolls two dices and tells you that there is at least one 6
  - What is the probability that the sum is at least 9?

- Conditional probability
  - If one rolls two dices and tells you that there is at least one 6
  - What is the probability that the sum is at least 9?
  - A = {there is at least one 6}, B = {sum is at leat 9}
  - A = {(1,6), (2,6), (3,6), (4,6), (5,6), (6,6), (6,1), (6,2), (6,3), (6,4), (6,5)}
  - P(A) = 11 / 36
  - A∩B = {(3,6), (4,6), (5,6), (6,6), (6,5), (6,4), (6,3)}
  - P(A∩B) = 7 / 36
  - P(B|A) = P(A∩B) / P(A) = 7 / 11

Conditional probability

- Multiplication rule for $P(A \cap B)$
  - $P(A|B) = \frac{P(A \cap B)}{P(B)}$
  - $P(A|B) * P(B) = \frac{P(A \cap B)}{P(B)} * P(B) \longrightarrow P(A \cap B) = P(A|B) * P(B)$

Conditional probability

- Multiplication rule for $P(A \cap B)$
  - A is the event that an individual having a particular disease
  - B is the event of negative blood test, P(B) = 0.25
  - The probability that the individual has the disease if his blood test is negative P(A|B) = 0.13
  - The probability that an individual with negative blood test has the disease is?

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Conditional probability

- Multiplication rule for P($A \cap B$)
  - A is the event that an individual having a particular disease
  - B is the event of negative blood test, P(B) = 0.25
  - The probability that the individual has the disease if his blood test is negative P(A|B) = 0.13
  - The probability that an individual with negative blood test has the disease is?
  - P(A∩ B) = P(A|B) * P(B) = 0.03

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Conditional probability

- The law of total probability
  - Exhaustive events (完备事件)
    - A_i are exhaustive events if one of them must occur
    - $\bigcup_i A_i = S$
  - If A_i are mutually exclusive and exhaustive events, for any other event B
  - P(B) = $\sum_i P(B|A_i) * P(A_i)$



Conditional probability

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  - If A_i are mutually exclusive and exhaustive events, for any other event B
  - P(B) = $\sum_i P(B|A_i) * P(A_i)$
    - $P(B|A_i) * P(A_i) = P(B \cap A_i)$
    - $\sum_i P(B|A_i) * P(A_i) = \sum_i P(B \cap A_i)$ = P(B)

Conditional probability

- Exercises
  - A certain manufacturer produces cars at two factories labeled A and B. 10% of the cars produced at factory A are found to be defective, while 5% of the cars produced at factory B are defective. Factory A produces 100,000 cars per year and factory B produces 50,000 cars per year
  - If a car from the manufacturer is defective, what is the probability that it is from factory A?

Conditional probability

- Exercises
  - A certain manufacturer produces cars at two factories labeled A and B. 10% of the cars produced at factory A are found to be defective, while 5% of the cars produced at factory B are defective. Factory A produces 100,000 cars per year and factory B produces 50,000 cars per year
  - If a car from the manufacturer is defective, what is the probability that it is from factory A?
  - A = {car is from A}, B = {car is from B}, D = {car is defective}
  - $P(A) = 100000 / 150000 = 2 / 3$, $P(B) = 50000 / 150000 = 1 / 3$, $P(D|A) = 0.1$, $P(D|B) = 0.05$
  - $P(D) = P(D|A) * P(A) + P(D|B) * P(B) = 0.08$
  - $P(A|D) = P(A \cap D) / P(D) = P(D|A) * P(A) / P(D) = 0.1 * 2 / 3 / 0.08 = 0.8$

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Conditional probability

- Exercises
  - A group of students consists of 60% male and 40% female. Among the male, 30% are foreign students, and among the female, 20% are foreign students
  - A student is randomly selected from the group and found to be a foreign student. What is the probability that the student is a female student?

Conditional probability

- Exercises
  - A group of students consists of 60% male and 40% female. Among the male, 30% are foreign students, and among the female, 20% are foreign students
  - A student is randomly selected from the group and found to be a foreign student. What is the probability that the student is a female student?
  - A = {the student is male}, B = {the student is female}, F = {the student is foreign student}
  - $P(A) = 60\%$, $P(B) = 40\%$, $P(F|A) = 30\%$, $P(F|B) = 20\%$
  - $P(F) = P(F|A) * P(A) + P(F|B) * P(B) = 30\% * 60\% + 20\% * 40\% = 0.18 + 0.08 = 0.26$
  - $P(B|F) = P(B \cap F) / P(F) = P(F|B) * P(B) / P(F) = 20\% * 40\% / 26\% = 30\%$

Conditional probability

- Baye`s theorem（贝叶斯定理）
  - If A_i are mutually exclusive and exhaustive events, for any other event B, the posterior probability （后验概率）of A_i given that B occurs
  - $P(A_i|B) = P(B|A_i) * P(A_i) \,/\, \sum_i P(B|\, A_i) * P(A_i)$
  - $P(A_i|B) = P(B \cap A_i) \,/\, P(B)$

Conditional probability

- Baye`s theorem
  - Only 1 in 1000 individuals can have a rare disease. A test is such that when an individual actually has the disease, a positive result will occur 99% of the time, whereas for an individual without the disease the test will show a positive result only 1% of the time
  - If a randomly selected individual is tested and the result is positive, what is the probability that the individual has the disease?

Conditional probability

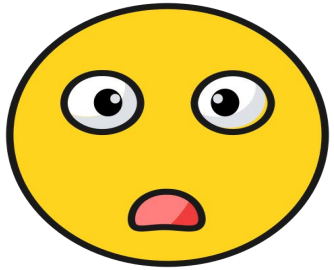
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  - If a randomly selected individual is tested and the result is positive, what is the probability that the individual has the disease?
    - If one has the disease, the test will be positive 99% of the time (true positive)
    - The test is very accurate, so if one has positive result, it seems that it is very likely for him to have the disease intuitively

Conditional probability

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  - Only 1 in 1000 individuals can have a rare disease. A test is such that when an individual actually has the disease, a positive result will occur 99% of the time, whereas for an individual without the disease the test will show a positive result only 1% of the time
  - If a randomly selected individual is tested and the result is positive, what is the probability that the individual has the disease?
    - Let A_1 = {individual has disease}, A_2 = {individual does not have disease}, B = {positive test result}
    - $P(A_1) = 0.001$, $P(A_2) = 0.999$, $P(B|A_1) = 0.99$, $P(B|A_2) = 0.01$
    - $P(A_1|B) = P(B|A_1) * P(A_1) / (P(B|A_1) * P(A_1) + P(B|A_2) * P(A_2)) = 0.09$
    - Even if the test is very accurate, when one has positive result the probability to have the disease is only 9% !

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    - Even if the test is very accurate, when one has positive result the probability to have the disease is only 9% !
    - Even though the test is 99% accurate, because the disease is so rare, the probability that one actually has the disease after testing positive is only about 9%
    - This counterintuitive result shows how important the prior probability is in Bayes' Theorem

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1. Sample pace and event
2. Axiom, interpretation, and property of probability
3. Counting technique
4. Conditional probability
5. Independance

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Independance

- Independant event (独立事件)
  - A and B are independent events, if the occurrence of one event has no effect on the occurrence of the other
  - $P(A|B) = P(A)$
  - $P(A \cap B) = P(A|B) * P(B) = P(B|A) * P(A)$
    $= P(A) * P(B) = P(B|A) * P(A) = P(B|A) = P(B)$
  - $P(B|A) = P(B)$
  - $P(A \cap B) = P(A) * P(B)$

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Independance

- Independant event (独立事件)
  - A work checks each product coming from a product line, and the probability for a prodcut to be defective is P(D), and not defective is P(N). If the worker checks *n* prodcuts
  - What is the probability that all of the products are defective?

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  - Can those two events independant? If yes then when?

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  - $P(A \cap B) = 0$, due to mutually exclusive
  - $P(A \cap B) = P(A) * P(B)$, due to the independance
  - So the only possible independance case is that the probability of at least one of them is 0

Independance

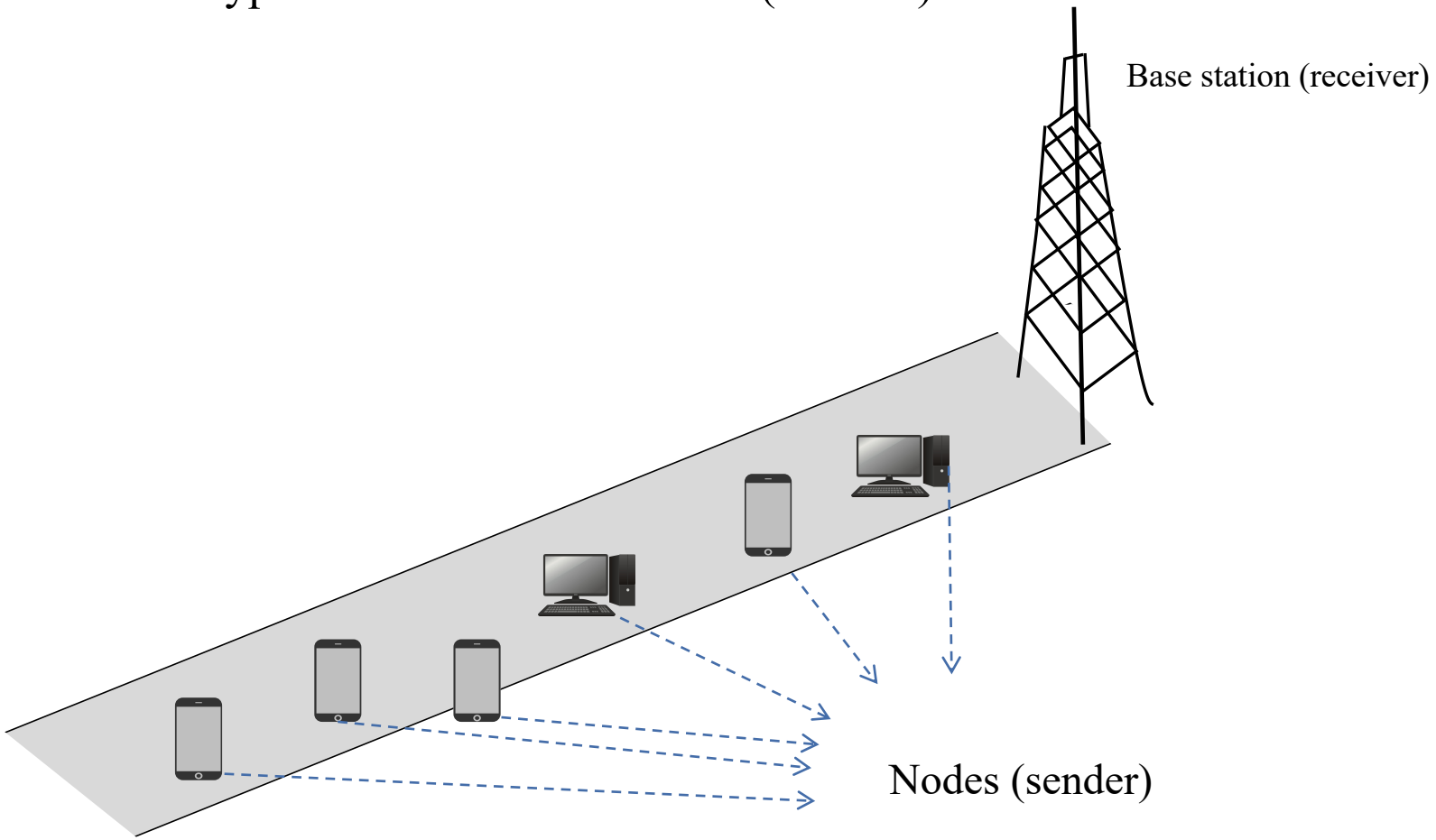
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  - So the only possible independance case is that the probability of at least one of them is 0
  - If the probability of each of them is larger than 0, it is impossible to be independant. Why?
  - In practice, mutually exclusive events are not independent because the occurrence of one event entirely rules out the occurrence of the other, which means the dependance

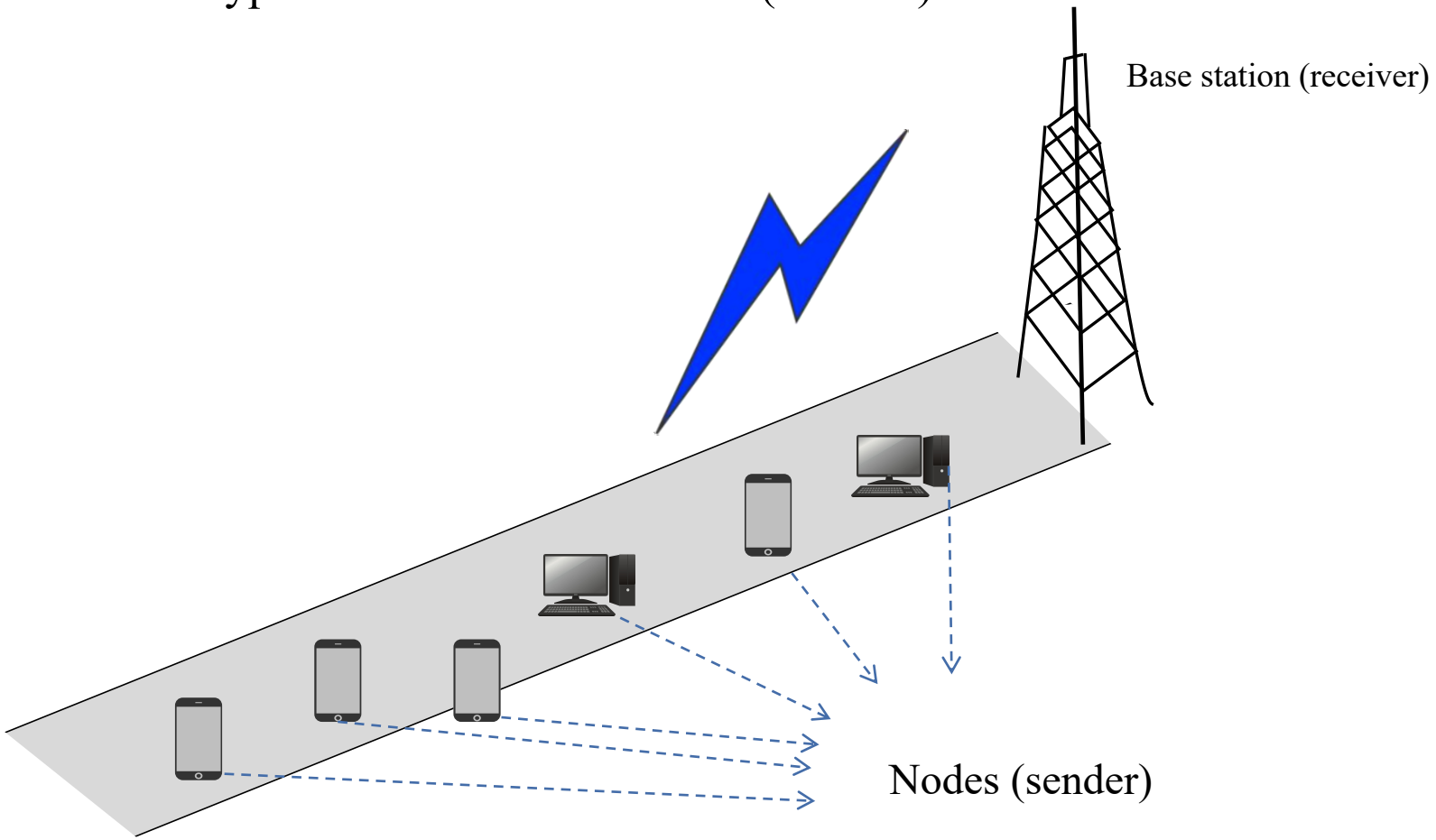
Independance

- Massive machine type communications of 6G (mMTC)



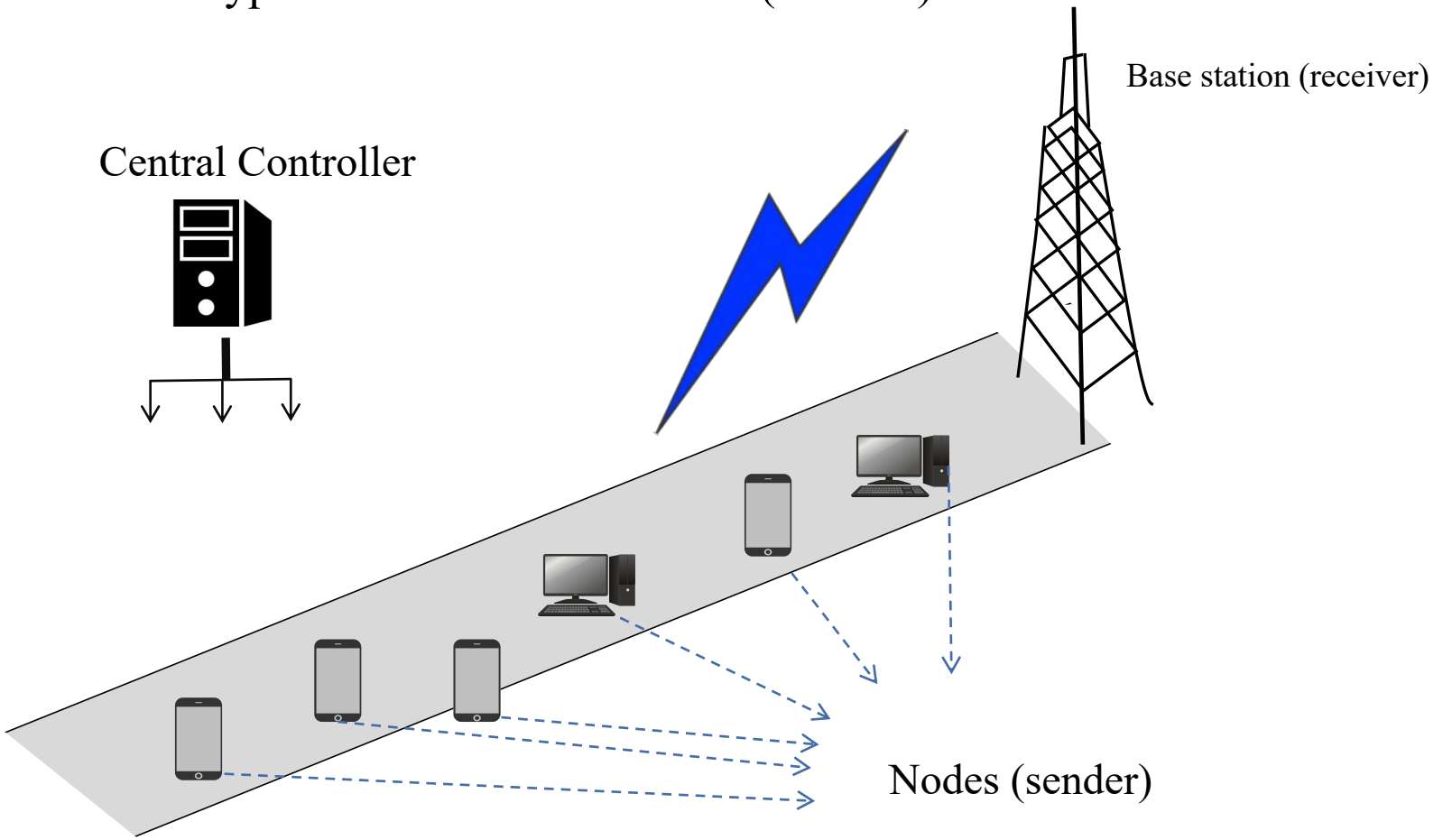
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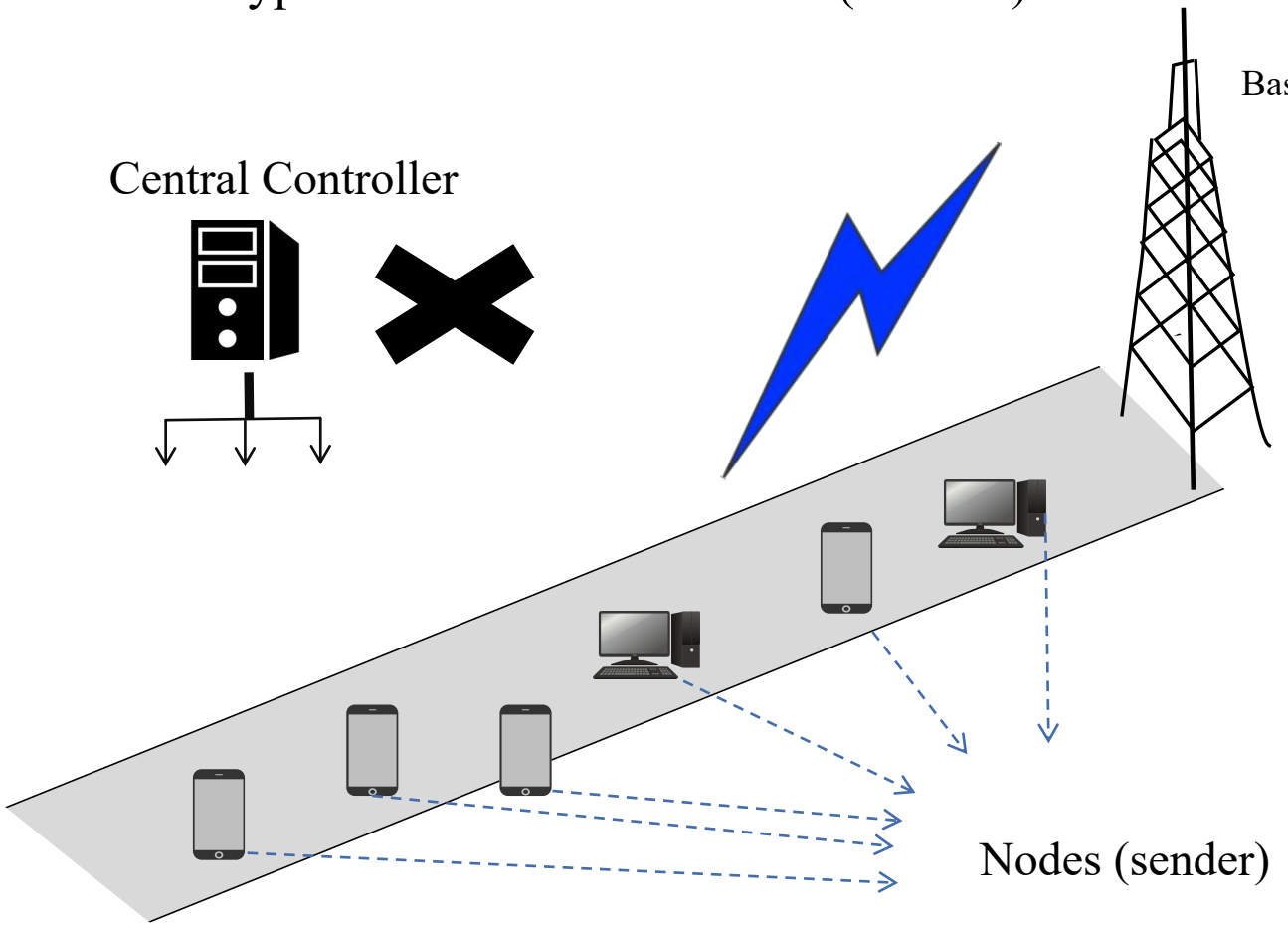
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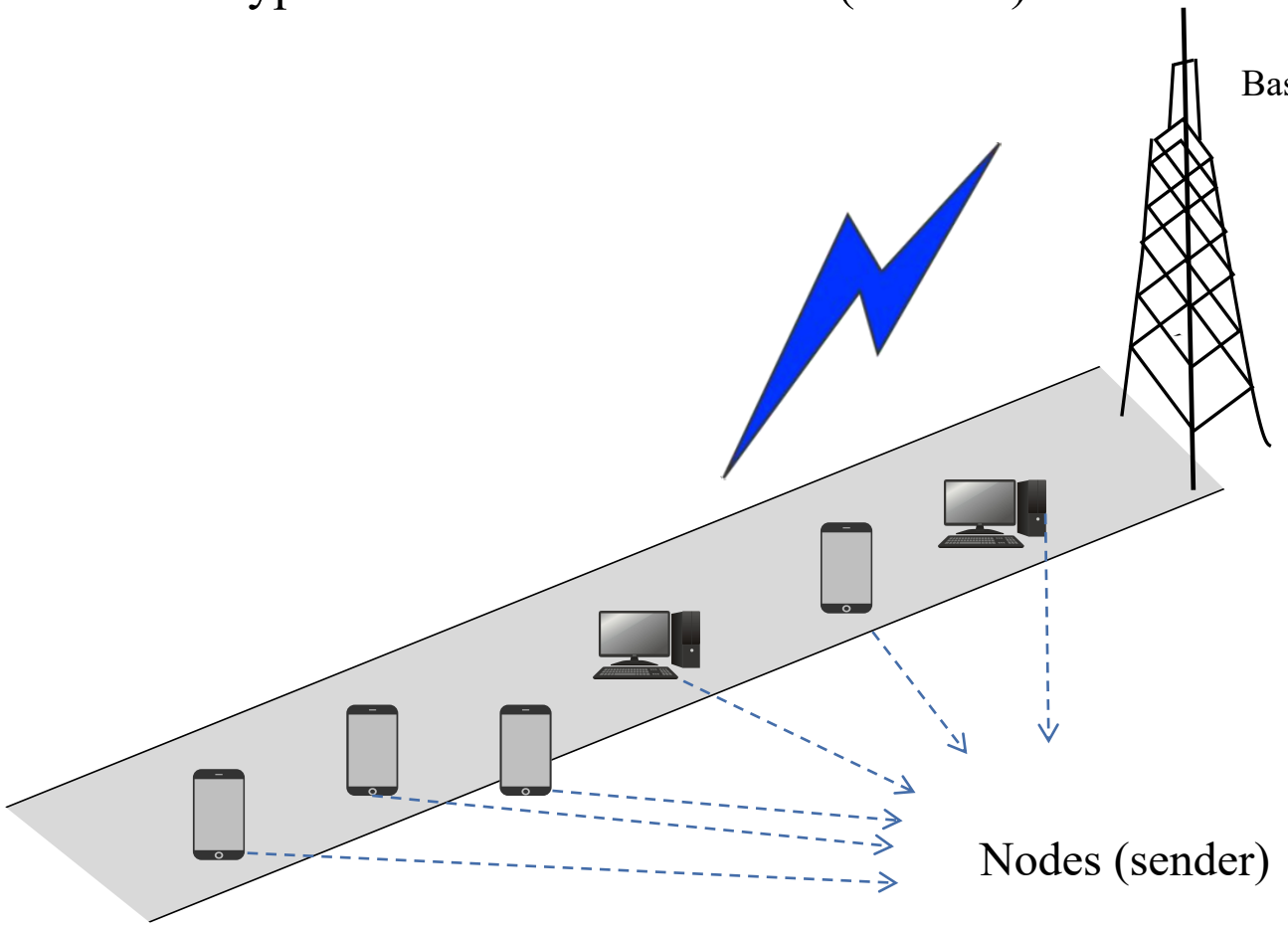
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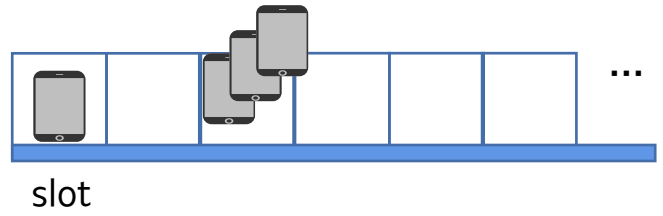
- **Huge number of nodes in 6G**
- **Central controlling is unavailable**
- **Random access**

Independance

- Massive machine type communications of 6G (mMTC)

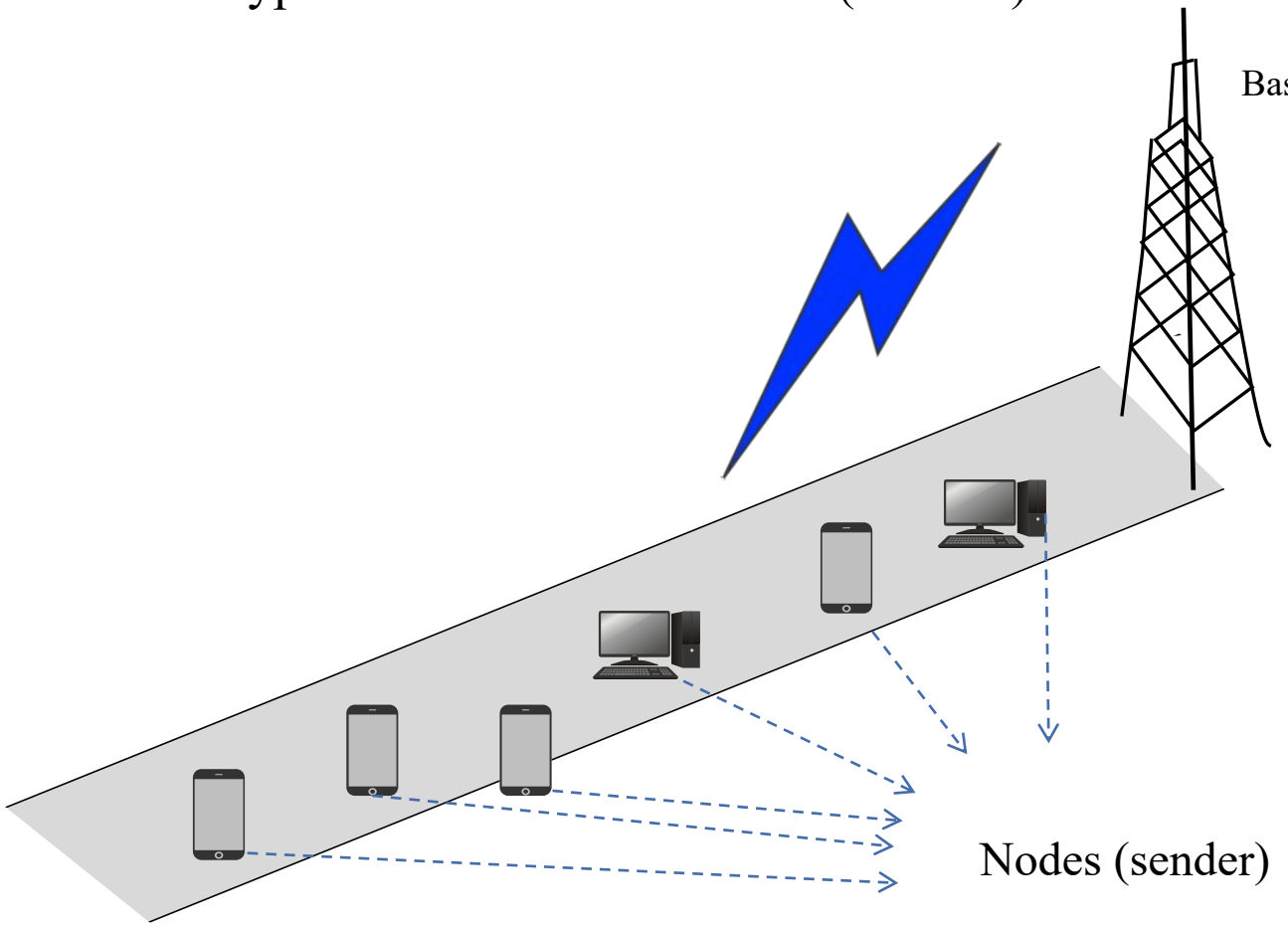


- **Time axis is divided into mini time slot**
- **In each time slot, each node independantly sends its signal based on a probability**

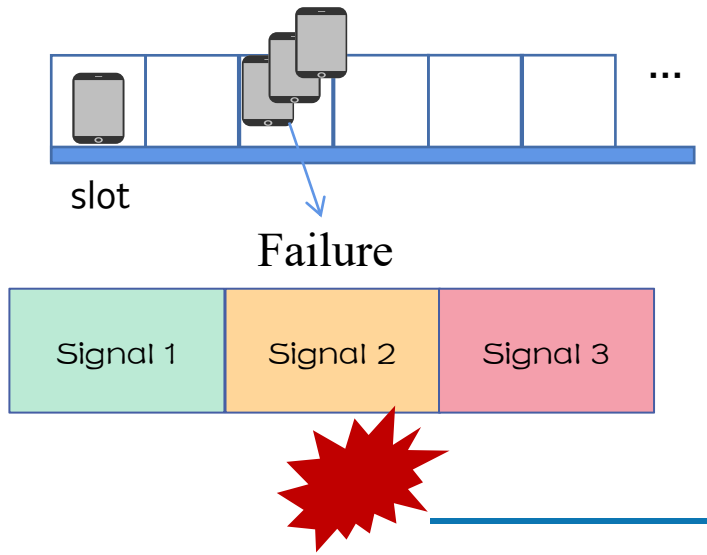


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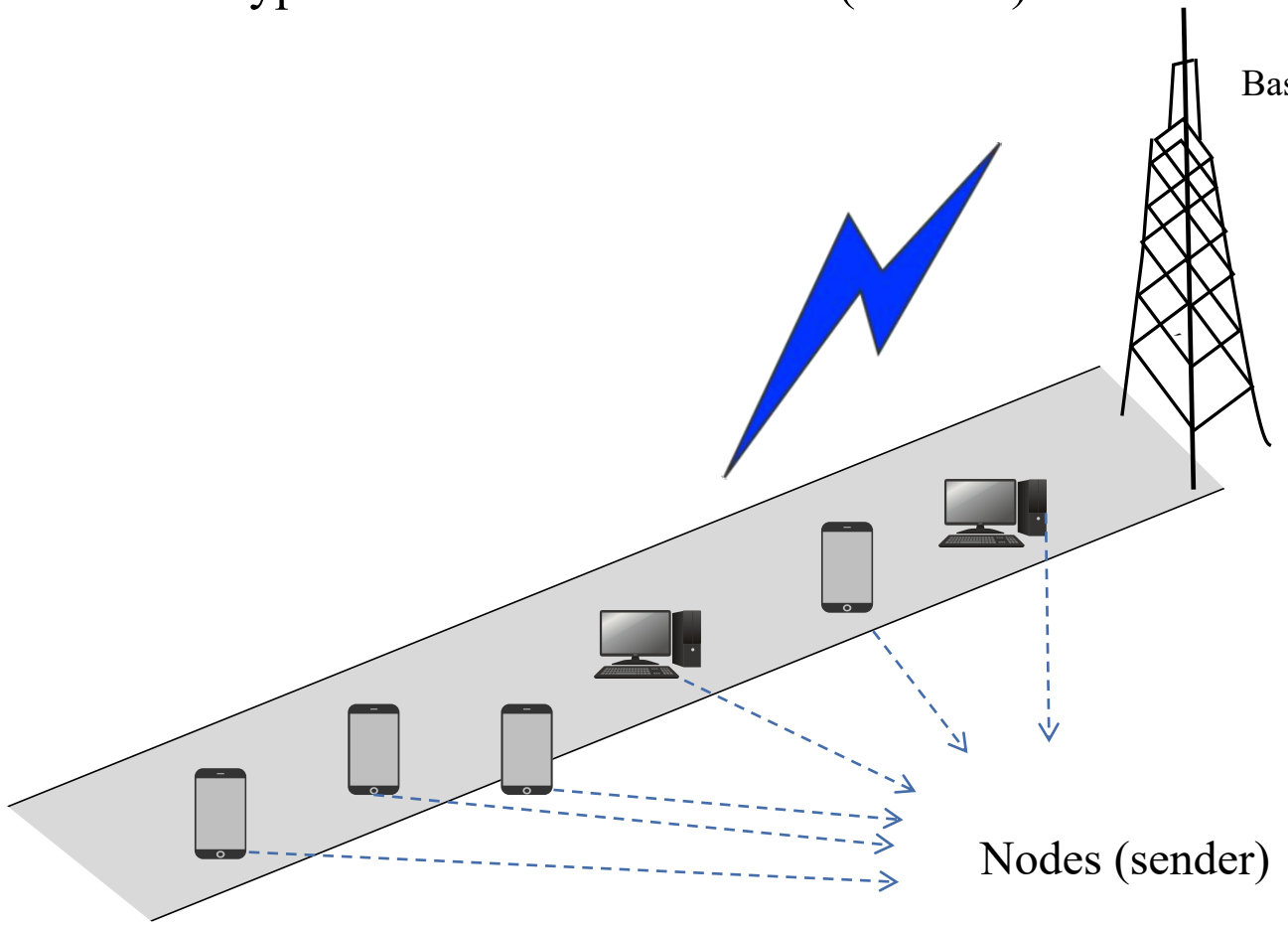


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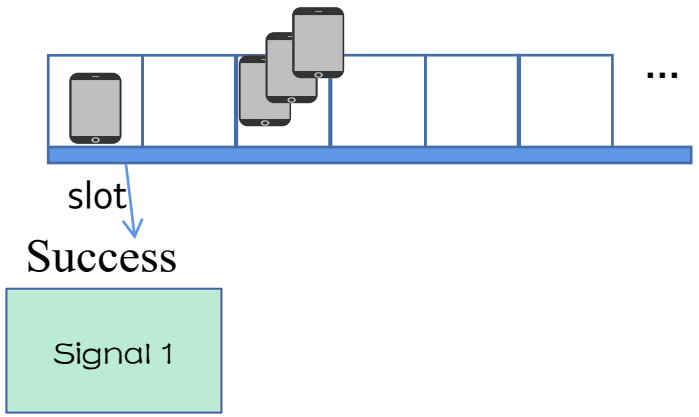


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Independance

- Massive machine type communications of 6G (mMTC)
- A successful transmission takes *s* seconds, a failure transmission takes *f* seconds, and idle slot takes *i* seconds
- Throughput of wireless network

- $\lim_{t\to\infty} \frac{\text{Number of successful transmissions}}{\text{t}}$

- If the number of nodes in the wireless network is *n*, the probability to transmit a signal in each slot is *p,* obtain the throughput

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- If the number of nodes in the wireless network is *n*, the probability to transmit a signal in each slot is *p,* obtain the throughput
- $\lim_{n\to\infty} \frac{\sum_{i=1}^{n} \mathbb{I}(\text{i})}{\sum_{i=1}^{n} length\ of\ i\ th\ slot}$
- $\mathbb{I}(\text{i}) = 1$ if in *i* th slot there is one successful transmission
- $\mathbb{I}(\text{i}) = 0$ otherwise
- $\lim_{n\to\infty} \frac{\sum_{i=1}^{n} \mathbb{I}(\text{i}) \ / \text{n}}{\sum_{i=1}^{n} length\ of\ i\ th\ slot\ /\ n} = \frac{Probability\ of\ success}{Average\ length\ of\ slot}$
- Probability of success = P(s) = $C_{1,n} * p * (1-p)^{n-1} = n * p * (1-p)^{n-1}$
- Probability of idle slot = P(i) = $(1-p)^{n}$
- Probability of failure = 1 - P(s) - P(i)
- Average length of slot = s * P(s) + i * P(i) + f * P(f)
- $\frac{Probability\ of\ success}{Average\ length\ of\ slot} = \frac{n*p*(1-p)^{n-1}}{s*P(s)+i*P(i)+f*P(f)}$

课后思考题

- Massive machine type communications of 6G (mMTC)
- There are two different types of node, A and B, the number of nodes A is n_1, B is n_2, transmission probability of A and B type are p_1 and p_2 respectively
- A successful transmission of type A and B node takes s_1 and s_2 seconds respectively, a failure transmission takes f seconds, and idle slot takes i seconds
- Throughput of wireless network?

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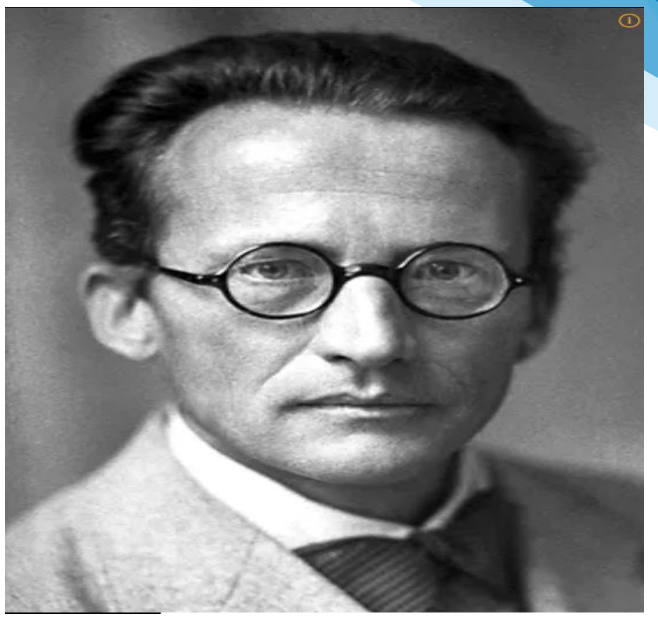
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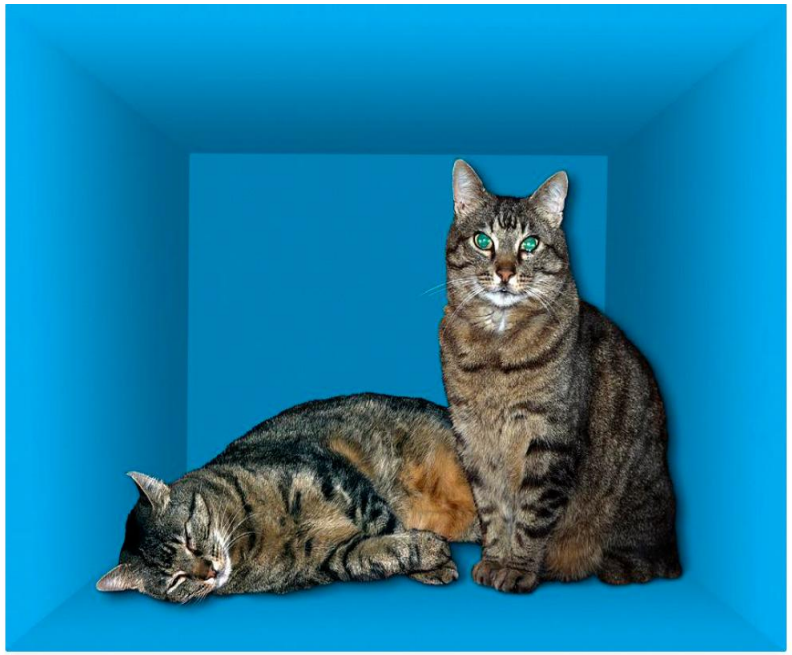
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- Throughput of wireless network?

- Throughput $= \frac{Probability\ of\ success}{Average\ length\ of\ slot}$
- $P(s_1) = C_{1,n1} * p_1 * (1-p_1)^{n1-1} * (1-p_2)^{n2}$
- $P(s_2) = C_{1,n2} * p_2 * (1-p_2)^{n2-1} * (1-p_1)^{n1}$
- P(s) = $P(s_1) + P(s_2)$
- P(i) = $(1-p_1)^{n1} * (1-p_2)^{n2}$
- P(f) = 1 - P(s) - P(i)
- Average length of slot =s_1 * P(s_1) + s_2 * P(s_2) + i * P(i) + f * P(f)
- Throughput $= \frac{P(s)}{Average\ length\ of\ slot}$

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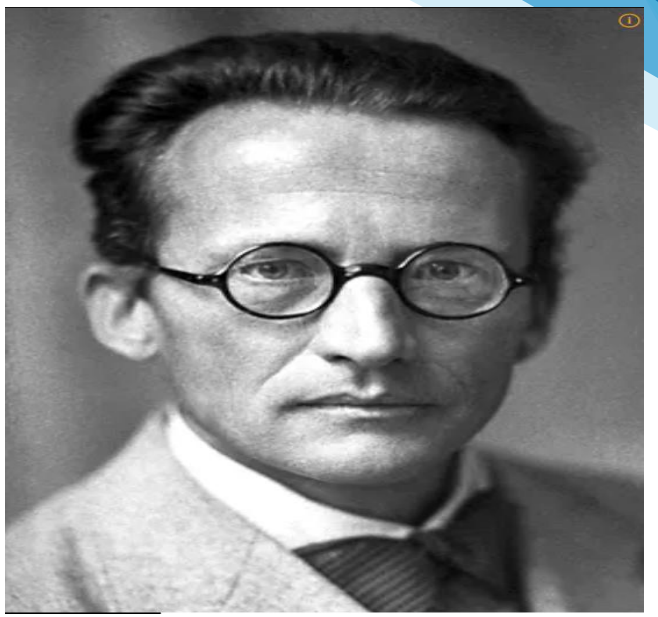
- Schrödinger's cat （薛定谔的猫）
  - 描述量子力学当中量子重叠的著名例子
  - 宏观世界的规律不适用于量子世界

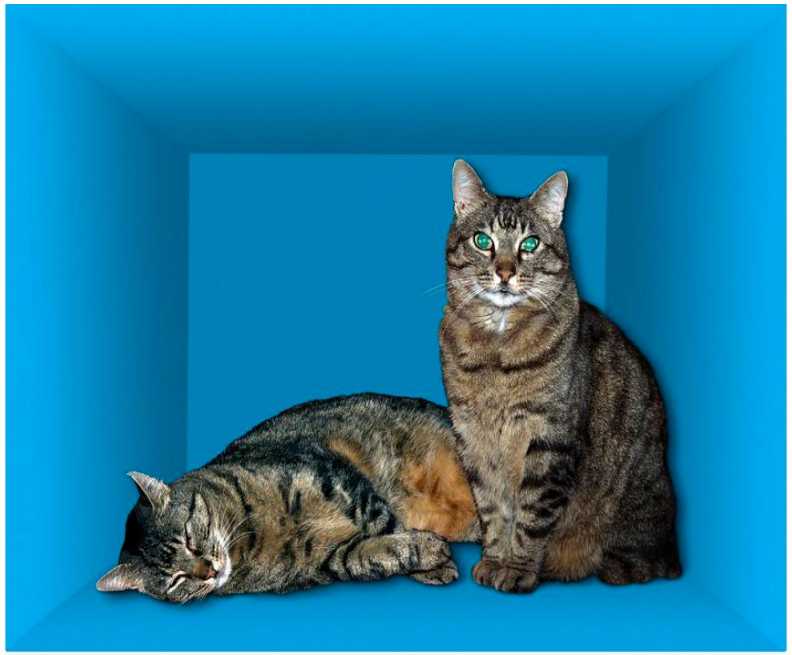




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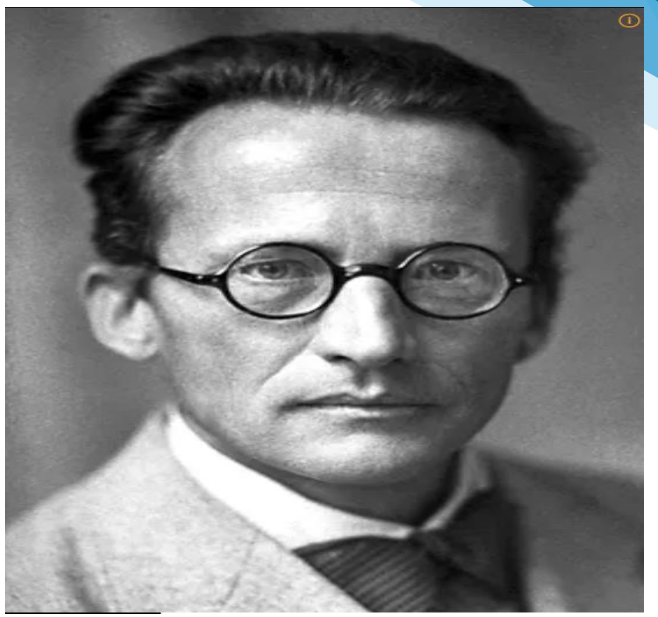
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  - 宏观世界
    - P(猫起初死了|观察到猫活着) = 0
    - P(猫起初活着|观察到猫活着) = 1

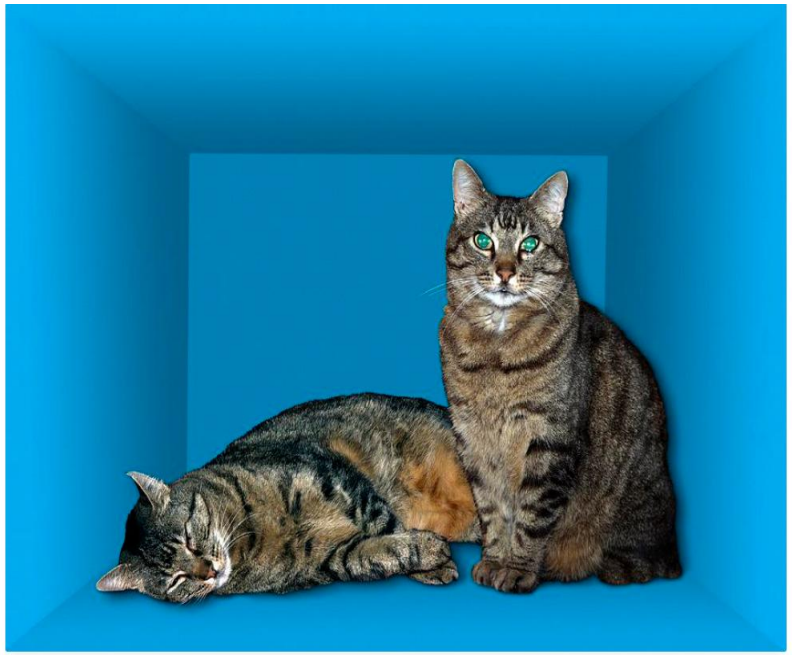




华南师范大學

- Schrödinger's cat （薛定谔的猫）
  - 描述量子力学当中量子重叠的著名例子
  - 宏观世界的规律不适用于量子世界
  - 宏观世界
    - P(猫起初死了|观察到猫活着) = 0
    - P(猫起初活着|观察到猫活着) = 1
  - 量子世界
    - P(猫起初死了|观察到猫活着) > 0
    - P(猫起初活着|观察到猫活着) > 0





華南師範大學

- Schrödinger's cat （薛定谔的猫）
  - 描述量子力学当中量子重叠的著名例子
  - 宏观世界的规律不适用于量子世界
  - 宏观世界
    - P(猫起初死了|观察到猫活着) = 0
    - P(猫起初活着|观察到猫活着) = 1
  - 量子世界
    - P(猫起初死了|观察到猫活着) > 0
    - P(猫起初活着|观察到猫活着) > 0
  - 在量子世界中，量子以多个形态同时存在，直到被观察为止

